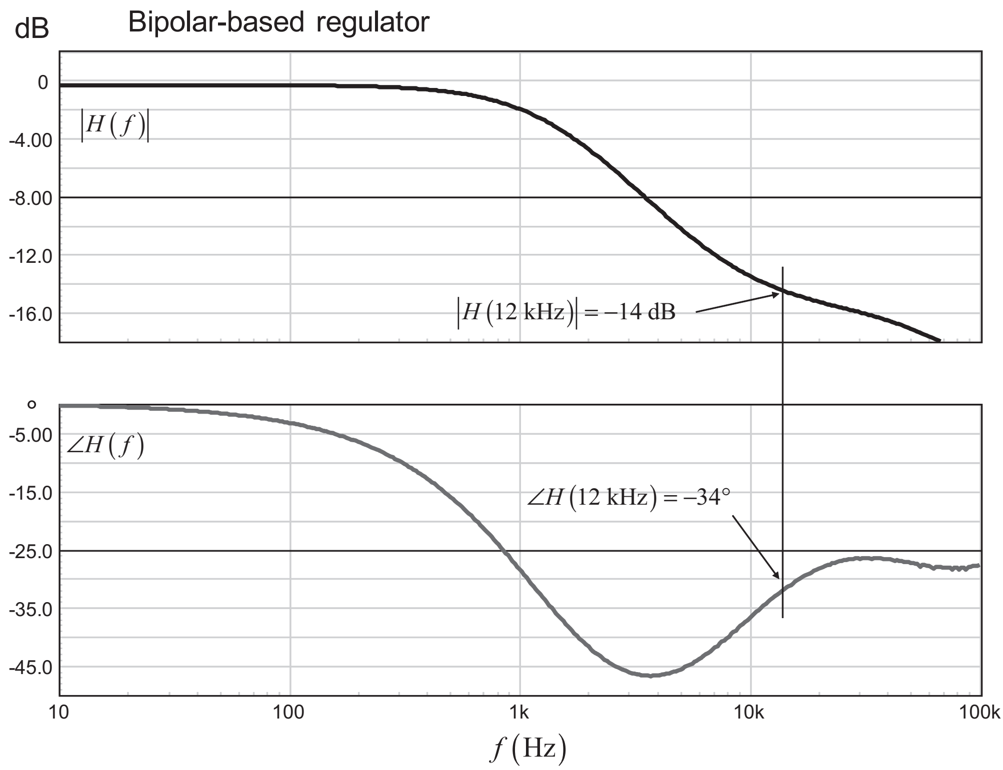

Figure 9.37 Ac results of the power stage show a gain deficit at the selected crossover frequency.

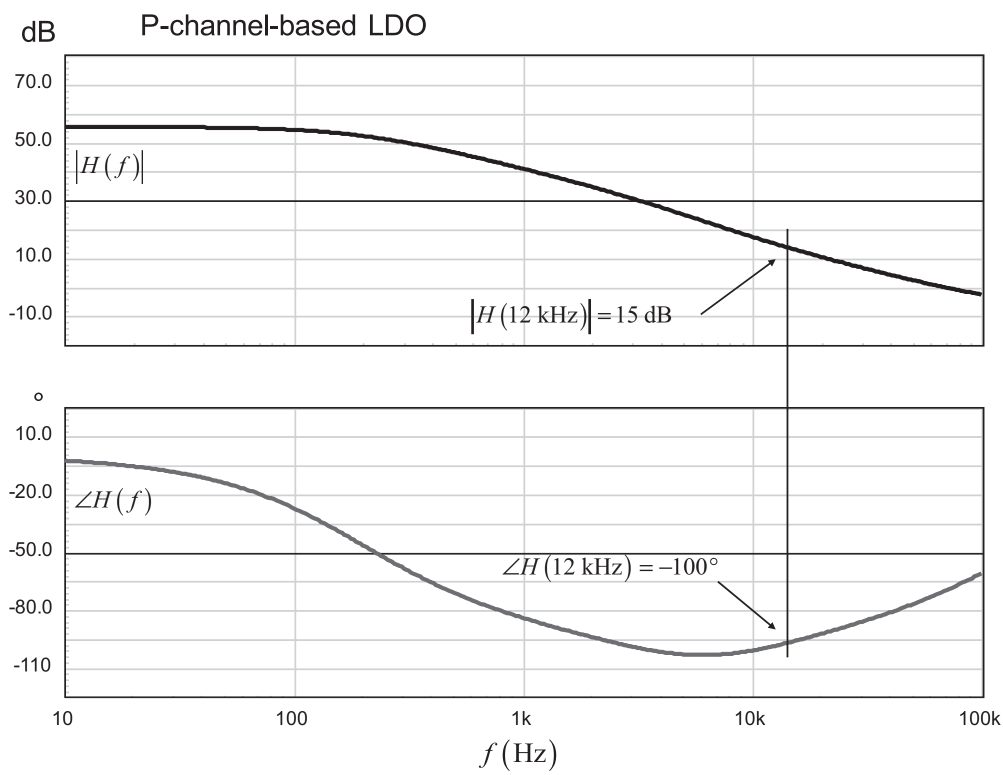

Figure 9.38 The power stage exhibits a large gain excess at the targeted crossover frequency value.

expresses a crossover value high enough so that the capacitive contribution becomes negligible, leaving the unavoidable ESR undershoot. This approximate formula is:

$$f_c \approx \frac{0.24}{C_{out} r_C} \tag{9.46}$$

In both cases, we have an output capacitor of 1 mF, affected by a 20-mΩ ESR value. According to (9.46), we must place the crossover frequency at:

$$f_c \approx \frac{0.24}{1000u \times 20m} = 12 \text{ kHz} \tag{9.47}$$

Extracting the transfer function data at this frequency gives us for the bipolar-based regulator:

$$\left|H(12 \text{ kHz})\right| = -14 \text{ dB} \tag{9.48}$$

$$\angle H(12 \text{ kHz}) = -34° \tag{9.49}$$

For the LDO type, we have:

$$\left|H(12 \text{ kHz})\right| = 15 \text{ dB} \tag{9.50}$$

$$\angle H(12 \text{ kHz}) = -100° \tag{9.51}$$

The lack of voltage gain for the first power stage makes sense since we drive a common-collector architecture. For the second, the inverter behind the op amp provides gain and the series-pass element does as well.

To check what type of compensator we need, let's evaluate the necessary phase boost at crossover for a 60° phase margin objective. In the first case, we have:

$$boost = \varphi_m - \arg T(f_c) - 90° = 60 + 34 - 90 = 4° \tag{9.52}$$

There is almost no phase boost required; the compensation could be realized with a simple type 1 compensator. For the LDO type, the necessary phase boost amounts to:

$$boost = \varphi_m - \arg T(f_c) - 90° = 60 + 100 - 90 = 70° \tag{9.53}$$

In this second case, we need a type 2 compensator. For the sake of comparing similar structures, we have adopted two type 2 compensators.

For the first regulator, given the 4° phase boost, both the pole and zero will be almost coincident:

$$f_p = \tan\left(\frac{boost}{2} + \frac{\pi}{4}\right) f_c = \tan\left(\frac{4}{2} + 45\right) \times 12k = 12.9 \text{ kHz} \tag{9.54}$$

$$f_z = \frac{f_c}{\tan\left(\frac{boost}{2} + \frac{\pi}{4}\right)} = \frac{12k}{\tan\left(\frac{4}{2} + 45\right)} = 11.2 \text{ kHz} \tag{9.55}$$

The mid-band gain is adjusted by the resistor R_2:

$$R_2 = \frac{R_4 f_p G}{f_p - f_z} \frac{\sqrt{\left(\frac{f_c}{f_p}\right)^2 + 1}}{\sqrt{\left(\frac{f_z}{f_c}\right)^2 + 1}} = \frac{10k \times 12.9k \times 10^{\frac{14}{20}}}{12.9k - 11.2k} \frac{\sqrt{\left(\frac{12k}{12.9k}\right)^2 + 1}}{\sqrt{\left(\frac{11.2k}{12k}\right)^2 + 1}} = 384 \text{ k}\Omega \tag{9.56}$$

R_4 in the previous formula is the upper resistor in the divider bridge.

The capacitors values come easily:

$$C_1 = \frac{1}{2\pi R_2 f_z} = 37 \text{ pF} \tag{9.57}$$

$$C_2 = \frac{C_1}{2\pi f_p C_1 R_2 - 1} = 247 \text{ pF} \tag{9.58}$$

You could also apply the k factor technique as described in Appendix 5B, as it would lead to the exact same results. These capacitors values are rather low, and, if necessary, it would be easy to increase them by changing the divider network impedance. We currently have 10 kΩ for R_4 and R_5. Should we diminish these two values to 1 kΩ, C_1 and C_2 would, respectively, equal 370 pF and 2.47 nF while R_2 would drop to 38 kΩ.

The compensation for the LDO follows a similar path. To provide a 70° phase boost at a 12-kHz crossover frequency, we will place a pole and a zero at:

$$f_p = \tan\left(\frac{boost}{2} + \frac{\pi}{4}\right) f_c = \tan\left(\frac{70}{2} + 45\right) \times 12k \approx 68 \text{ kHz} \tag{9.59}$$

$$f_z = \frac{f_c}{\tan\left(\frac{boost}{2} + \frac{\pi}{4}\right)} = \frac{12k}{\tan\left(\frac{70}{2} + 45\right)} = 2.1 \text{ kHz} \tag{9.60}$$

The mid-band gain is adjusted by the resistor R_2:

$$R_2 = \frac{R_4 f_p G}{f_p - f_z} \frac{\sqrt{\left(\frac{f_c}{f_p}\right)^2 + 1}}{\sqrt{\left(\frac{f_z}{f_c}\right)^2 + 1}} = \frac{10k \times 68k \times 10^{-\frac{15}{20}}}{68k - 2.1k} \frac{\sqrt{\left(\frac{12k}{68k}\right)^2 + 1}}{\sqrt{\left(\frac{2.1k}{12k}\right)^2 + 1}} = 1.8 \text{ k}\Omega \tag{9.61}$$

R_4 in the previous formula is the upper resistor in the divider bridge.

The capacitors values come easily:

$$C_1 = \frac{1}{2\pi R_2 f_z} = 49 \text{ nF} \tag{9.62}$$

$$C_2 = \frac{C_1}{2\pi f_p C_1 R_2 - 1} = 1.3 \text{ nF} \tag{9.63}$$

We can now run another ac simulation and check the loop gain response by probing the error amp voltage. The results for both configurations appear in Figure 9.39 and are very close to what we wanted. Some discrepancy can always happen, especially if we consider the op amp origin pole role.

9.3.3 Testing the Transient Response

We now have all of our component values. They are passed to the simulator as simple values or they can be automated as shown on the graph. Automation is interesting if you want to explore different phase margins on the fly: the corresponding transient response is immediate as all elements are recomputed before simulation start. To test our regulator step load response, we will sweep the output current from 1 A to 2 A in 10 μs. We purposely remove the inductive contribution to precisely see the resistive effect. The results appear in Figure 9.40.

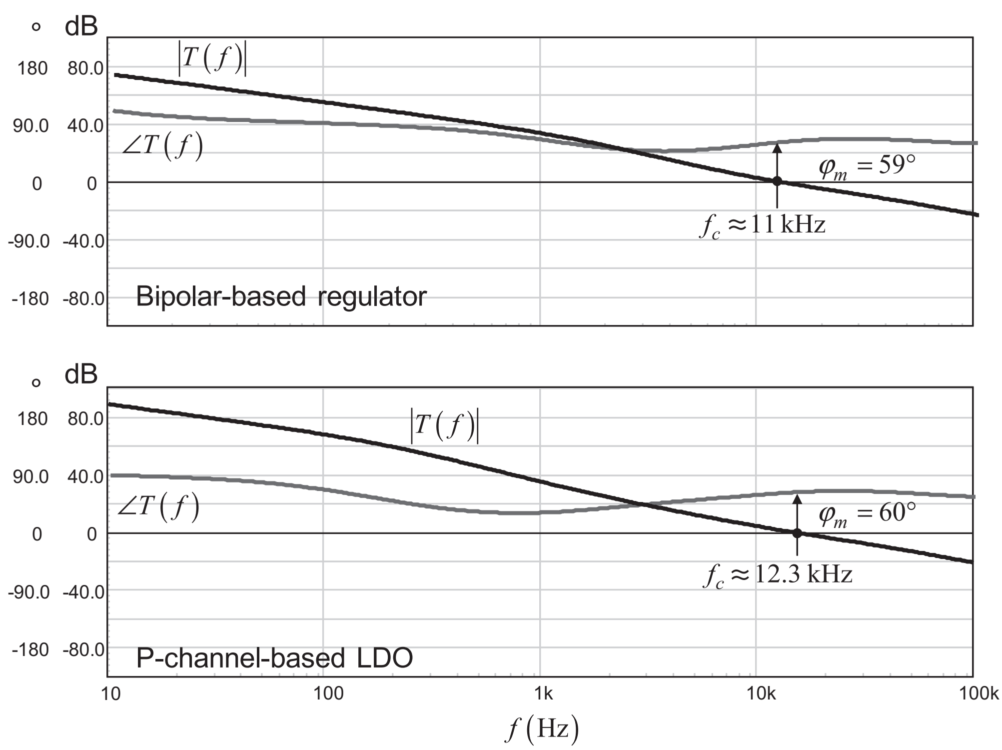

Figure 9.39 The ac responses show the right crossover frequencies and phase margins on both designs.

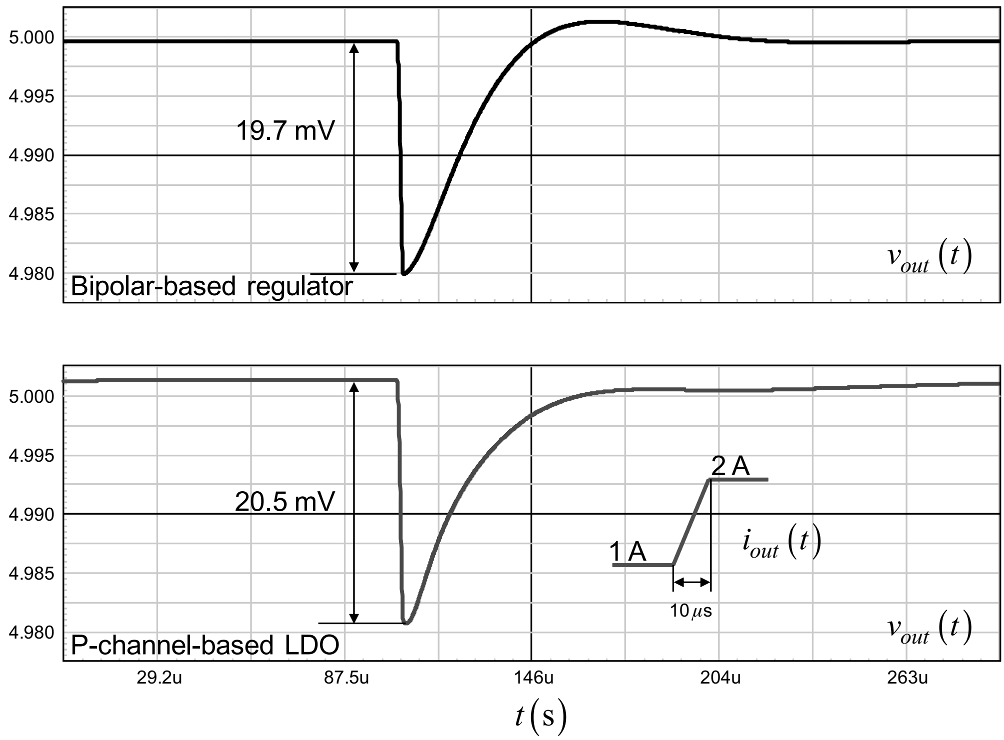

Figure 9.40 The transient response shows that the sole contributor to the voltage undershoot is the capacitor ESR.

The upper portion of the graph shows the bipolar-based regulator response. The compensation is performed by a pure integrator (pole and zero are almost coincident, turning the type 2 into a type 1). As expected, there is slight overshoot (0.03 percent), but the undershoot is exactly dictated by the ESR. The capacitive contribution does not exist. The following graph shows the response of the LDO for the same output stimulus. In this case, we have a zero located at 2 kHz. The recovery time is a bit longer (because of the zero presence in the low frequency portion) but the slight overshoot is gone. The undershoot is nicely given by the ESR alone, no capacitive contribution.

For the sake of the example, if we decrease the crossover frequency to 5 kHz we should see the capacitive contribution coming back in the picture. Figure 9.41 demonstrates this fact with the P-channel-based LDO. The whole response is no longer given by the ESR alone, but by the capacitor as well. As the crossover went down, the response time naturally expands.

9.4 Design Example 3: A CCM Voltage-Mode Boost Converter

Among switching converters, the boost topology is a popular design whose applications range from batteries-operated portable devices to high power ac-dc preconverters. Using two storage elements, this second-order converter can be designed to operate in the CCM at the lowest input voltage and the heaviest load. This design

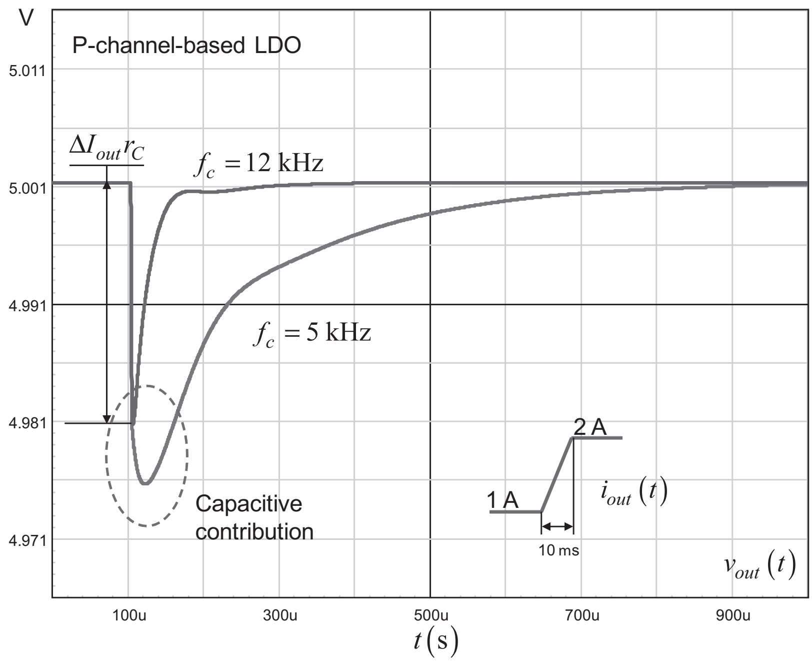

Figure 9.41 When the crossover frequency decreases to 5 kHz, the capacitive contribution shows up again.

example shows how to stabilize a CCM voltage-mode boost converter delivering 19 V from a car battery.

9.4.1 The Power Stage Transfer Function

You know the compensation of a given converter starts with its small-signal transfer function noted $H(s)$. Within this transfer function appear multiple poles, zeros, static gains, and so on. The dc operating point of the converter represents an important starting point which is, most of the time, required to compute the position of the aforementioned poles and zeros. With a boost converter operated in CCM and implementing voltage mode control, the dc transfer function M, when neglecting ohmic losses, is defined by:

$$M = \frac{V_{out}}{V_{in}} = \frac{1}{1-D} \tag{9.64}$$

where D represents the duty ratio that can be immediately extracted from (9.64):

$$D = \frac{V_{out} - V_{in}}{V_{out}} \tag{9.65}$$

The CCM boost transfer function appears in numerous textbooks and [12] has compiled and documented several other topologies in both voltage and current-mode controls. The control-to-output transfer function of the CCM boost converter operating in voltage mode appears in the following equation:

$$\frac{V_{out}(s)}{V_{err}(s)} = H_0 \frac{\left(1+\frac{s}{\omega_{z_1}}\right)\left(1-\frac{s}{\omega_{z_2}}\right)}{1+\frac{s}{Q\omega_0}+\left(\frac{s}{\omega_0}\right)^2} \tag{9.66}$$

By control-to-output, we mean how the ac voltage present on the error amplifier output propagates through the pulse width modulator (PWM) and drives the boost converter output voltage. Equation (9.66) indicates the presence of two zeros, one of them being our right half plane zero (RHPZ, ω_{z2}) plus a double pole located at ω_0, affected by a quality factor Q. The definitions of these elements are the following:

$$\omega_{z_1} = \frac{1}{r_C C} \tag{9.67}$$

$$\omega_{z_2} \approx \frac{R(1-D)^2}{L} \tag{9.68}$$

$$H_0 = \frac{{V_{out}}^2}{V_{in}V_{peak}} \tag{9.69}$$

$$\omega_0 \approx \frac{1-D}{\sqrt{LC}} \tag{9.70}$$

$$Q \approx \frac{\omega_0}{\frac{r_L}{L}+\frac{1}{C(r_C+R)}} \tag{9.71}$$

where r_C represents the output capacitor C ESR, V_{in} and V_{out} are, respectively, the input and output voltages; V_{peak} is the PWM modulator sawtooth amplitude; L is the boost converter inductor affected by r_L, its series resistance; D is the converter duty ratio; and Q the quality factor linked to the double pole.

To illustrate the boost converter ac response, we will take the example of a simple 60-W dc-dc converter elevating a 12-V battery voltage to a 19-V level, as in a notebook application powered from a car battery. Its specifications are as follows:

$V_{in,max} = 15$ V
$V_{in,min} = 11.5$ V
$V_{out} = 19$ V
$I_{out} = 3$ A
$R = 19/3 = 6.33\ \Omega$

Phase margin at cross over greater or equal to 60°, worse case.

The calculation steps using software from [17] give us the following values:

F_{sw} = 100 kHz
V_{peak} = 2 V
L = 50 μH, r_L = 10 mΩ
C = 1000 μF, r_C = 20 mΩ

From these values, we first evaluate the operating duty cycle at the lowest input voltage thanks to (9.65):

$$D = \frac{19 - 11.5}{19} = 0.395 \tag{9.72}$$

Then, we can proceed with the poles/zeros positions as defined by equations (9.67) through (9.71):

$$f_{z_1} = \frac{1}{2\pi \times 20m \times 1m} = 7.9 \text{ kHz} \tag{9.73}$$

$$f_{z_2} = \frac{6.33 \times (1 - 0.395)^2}{2\pi \times 50u} = 7.4 \text{ kHz} \tag{9.74}$$

$$20\text{Log}_{10}(H_0) = 20\text{Log}_{10}\left(\frac{19^2}{11.5 \times 2}\right) = 23.9 \text{ dB} \tag{9.75}$$

$$f_0 = \frac{1 - 0.395}{2\pi \times \sqrt{50u \times 1m}} = 430.8 \text{ Hz} \tag{9.76}$$

$$Q = \frac{2.7k}{\dfrac{10m}{50u} + \dfrac{1}{1m \times (20m + 6.33)}} = 7.57 \text{ or } 17.6 \text{ dB} \tag{9.77}$$

These results indicate the presence of a double pole placed at 430 Hz and affected by a 17.6-dB quality factor. This resonant frequency will move in relationship to the input voltage as it affects the duty ratio. The worst case occurs at the minimum input voltage and full load. However, once the compensation circuit is calculated, it is of the designer duty to make sure that the phase and gain margins do not degrade over the whole input voltage and load ranges. Also, as parasitic elements are involved, their impact on the final loop gain must also be carefully evaluated.

Once we have the numerical results, we have several choices to plot the CCM boost converter Bode response. One possibility consists of using scientific software such as Mathcad®, which can directly handle modules and imaginary notation via dedicated functions. This is the simplest and fastest approach. Unfortunately, for people who do not own such a software, we need to offer something simpler. For instance, directly extracting the module and argument of (9.66) via the poles and zeros positions is a possible solution that can be easily used with Excel® for instance. Capitalizing on this remark, the module and argument of $H(s)$ appear through (9.78) and (9.79).

$$|H(f)| = 20\text{Log}_{10}\left[H_0 \frac{\sqrt{1+\left(\frac{f}{f_{z1}}\right)^2}\sqrt{1+\left(\frac{f}{f_{z2}}\right)^2}}{\sqrt{\left(1-\left(\frac{f}{f_0}\right)\right)^2+\left(\frac{f}{f_0 Q}\right)^2}}\right] \tag{9.78}$$

$$\arg H(f) = \tan^{-1}\left(\frac{f}{f_{z1}}\right) - \tan^{-1}\left(\frac{f}{f_{z2}}\right) - \tan^{-1}\left[\frac{f}{f_0 Q}\frac{1}{1-\left(\frac{f}{f_0}\right)^2}\right] \tag{9.79}$$

Once these equations are entered into the calculation tool of your choice, the complete Bode diagram can be unveiled, as shown in Figure 9.42.

We are now to the point where you need to select the crossover frequency. The crossover frequency of a CCM boost converter is limited in the upper range by the lowest RHPZ position. As explained in Chapter 2, we need to limit the duty ratio slew-rate by selecting a crossover frequency well below the worse case RHPZ position. Experience shows that below 30 percent of this position gives adequate results. In our case, based on what (9.74) tells us, the maximum crossover frequency we can reasonably obtain is:

$$f_c < 0.3 \times 7.4k < 2.2 \text{ kHz} \tag{9.80}$$

Another consideration concerns the resonant frequency. You must always consider a crossover frequency beyond the region where the peak occurs. Otherwise, the loop will not offer enough gain to damp the LC network peaking in the output impedance expression, bringing instability to your converter. A common rule is

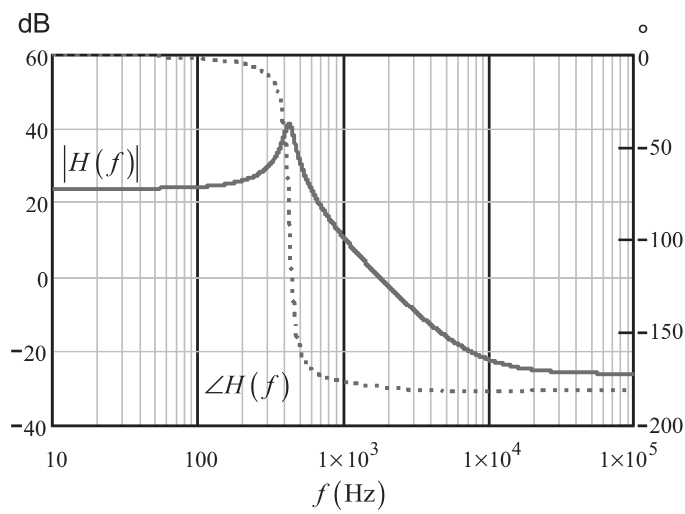

Figure 9.42 The boost converter CCM transfer function can quickly be plotted thanks to (9.78) and (9.79).

to recommend at least three times the maximum resonant frequency. In the CCM boost converter, this resonant frequency changes with the input conditions, reaching its maximum for a 15-V input level. In this case, (9.76) predicts a resonant peak placed at 562 Hz. Therefore, the crossover frequency must stay above:

$$f_c > 3 \times 562 > 1.68 \text{ kHz} \tag{9.81}$$

Combining (9.80) and (9.81), we will adopt a crossover point of 2 kHz. What if (9.81) would give an unrealistically high value? In that case, you would still have the solution to increase the output capacitor to push the resonance in a lower frequency region. Nevertheless, you must still make sure that the selected crossover frequency and the available capacitor lead to a theoretical undershoot compatible with your specification. Otherwise, another calculation iteration is necessary.

Now that we have selected a crossover frequency, you can either read the graph or use the magnitude/phase equations to extract the module and the phase rotation at 2 kHz for a maximum load and a 11-V input voltage. We obtain

$$|H(2k)| = -1.77 \text{ dB} \tag{9.82}$$

$$\arg H(2k) = -179° \tag{9.83}$$

9.4.2 Compensating the Converter

The loop stability exercise consists of shaping the compensator $G(s)$ to provide a gain at 2 kHz that compensates the gain deficiency (or excess) extracted at the crossover frequency. This way, we will satisfy $|H(f_c)G(f_c)| = 1$ implying a crossover at f_c. In our example, we must provide a gain of 1.77 dB when the frequency reaches 2 kHz. For the phase, it is a bit different. You need to provide a certain amount of excess phase at the crossover frequency to obtain the required phase margin *PM*. If we look for a 60° phase margin, this *boost* is calculated as follows:

$$boost = \varphi_m - \arg H(f_c) - 90 = 60 + 179 - 90 = 149° \tag{9.84}$$

The phase boost is obtained by placing poles and zeros at proper places. For a boost converter operating in CCM and a needed phase boost greater than 90°, you must use a type 3 compensator. Such an amplifier appears in Figure 9.43, together with the boost power stage configuration.

How do we adjust the elements to provide the necessary gain and where to place the poles and zeros? If the *k* factor works reasonably well for a first-order converter, it often leads to conditional stability in the case of a CCM converter featuring a resonant double pole. This is because the method solely focuses on the crossover region and not what happened before or beyond. For this reason, the manual placement is the preferred way of compensating when complex Bode plots

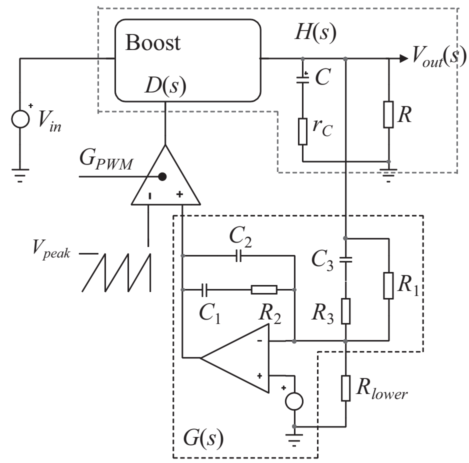

Figure 9.43 A type 3 compensator can be built with an operational amplifier. It can theoretically boost the phase up to 180°.

are in play. Before we present the method, we first need to introduce the transfer function of the type 3 amplifier. It can be shown that it looks as follows:

$$|G(f)| \approx 20 \cdot \mathrm{Log}_{10}\left[\frac{R_2}{R_3}\frac{\sqrt{1+\left(\frac{f_{z1}}{f}\right)^2}\sqrt{1+\left(\frac{f_{z2}}{f}\right)^2}}{\sqrt{1+\left(\frac{f}{f_{p1}}\right)^2}\sqrt{1+\left(\frac{f_{p2}}{f}\right)^2}}\right] \tag{9.85}$$

It brings a pole at the origin, a double zero, and a double pole. The pole placed at the origin is found on the vast majority of compensators. It is there to provide high gain in the dc region. This high dc gain reduces the output impedance and improves the static error and the input rejection. This origin pole naturally lags the phase by 90° (π/2) to which you must add the 180° inversion brought by the op amp (π). Capitalizing on these remarks, the *total* argument of the type 3 arrangement appears as described by (9.86).

$$\arg G(f) = \pi - \left(\tan^{-1}\frac{f}{f_{z1}} + \tan^{-1}\frac{f}{f_{z2}} - \tan^{-1}\frac{f}{f_{p1}} - \tan^{-1}\frac{f}{f_{p2}} - \frac{\pi}{2}\right) \tag{9.86}$$

The most troublesome contributor in the CCM boost converter transfer function is the double pole located at the resonant frequency. One possible solution is to place in G a double zero right at this location. In some converters, it is also interesting to split these zeros, one placed at the resonant frequency and the second slightly below it, to increase the phase margin in the DCM.

If the ESR zero occurs before the crossover frequency, we place a pole right at its location to force the gain decrease. The second pole will be placed at half the switching frequency to make sure enough gain margin exists as the phase margin vanishes to 0°. In our case, the ESR zero occurs after the crossover frequency, but not too far away from it. To show the impact of this first pole position in the compensator, we will cover two possible strategies.

Strategy 1

1. A double zero f_{z1} and f_{z2} is placed at the lowest resonant frequency (e.g., 430 Hz).
2. A first pole f_{p1} is placed right at the ESR zero occurrence, 7.9 kHz. The purpose of this pole is to force a –1 slope despite the presence of the boost converter resonant pole cancelled by the double zero.
3. A second pole f_{p2} will be placed at half of the switching frequency (50 kHz) to ensure a proper gain rolloff as the phase rotation further degrades. This pole gives the necessary gain margin we need (at least 10–15 dB).
4. Once all locations are known, the resistor R_2 from (9.85) is extracted to provide the proper gain compensation at f_c.
5. Given the selection of the poles/zeros affecting $G(s)$, we will evaluate the loop $T(s)$ resulting phase margin and see if it complies with our specification.

Strategy 2

1. A double zero f_{z1} and f_{z2} is placed below the lowest resonant frequency (e.g., 300 Hz).
2. A first pole f_{p1} is now computed to provide the required phase margin (e.g., 60°).
3. A second pole f_{p2} will be placed at half of the switching frequency (50 kHz) to ensure a proper gain rolloff as the phase rotation further degrades.
4. Once all locations are known, the resistor R_2 from (9.85) is calculated to provide the proper gain compensation at f_c.

When $H(s)$ and $G(s)$ are combined together, we observe the total loop noted $T(s)$. The loop phase margin PM after compensation is evaluated as the distance of the total phase rotation at f_c i.e. arg $T(f_c)$ and the 0° limit. Mathematically, it can be expressed as follows:

$$\varphi_m = \arg H(f_c) + \arg G(f_c) - 0° \tag{9.87}$$

Strategy 1 applied to (9.86) gives us a total phase lag for the compensator G at 2 kHz equal to:

$$\arg G(f) = 180 + 2\tan^{-1}\frac{2k}{430} - \tan^{-1}\frac{2k}{7.9k} - \tan^{-1}\frac{2k}{50k} - 90 \approx 229° \tag{9.88}$$

From (9.87), it leads to a final loop phase margin of:

$$\varphi_m = \arg H(f_c) + \arg G(f_c) = -179 + 229 = 50° \tag{9.89}$$

with 50° at an 11.5-V input voltage, increasing to 56° when the boost converter is supplied from 15 V.

For strategy number 2, we will fix the double zero to 300 Hz and adjust the position of the first pole to cope with our 60° phase margin requirement. We know from (9.84) that if the compensator two zeros and the second pole are fixed, the first pole can be placed to offer the required boost of 149°. In (9.86), the equation includes the origin pole phase lag plus the op amp phase reversal. However, the real boost brought by the compensator G is calculated as the phase bump amplitude above the –270° or –3π/2 limit (see Figure 9.44). Thus, from (9.86), we can extract the contribution of the first pole f_{p1} by removing the –3π/2 term. Therefore, we have:

$$\tan^{-1}\left(\frac{f_c}{f_{p1}}\right)=\tan^{-1}\left(\frac{f_c}{f_{z1}}\right)+\tan^{-1}\left(\frac{f_c}{f_{z2}}\right)-\tan^{-1}\left(\frac{f_c}{f_{p2}}\right)-boost \tag{9.90}$$

The pole position is now easily derived and follows (9.91):

$$f_{p1}=\frac{f_c}{\tan\left[\tan^{-1}\left(\frac{f_c}{f_{z1}}\right)+\tan^{-1}\left(\frac{f_c}{f_{z2}}\right)-\tan^{-1}\left(\frac{f_c}{f_{p2}}\right)-boost\right]}=9.9\text{ kHz} \tag{9.91}$$

Using this second strategy, we can recompute our phase margin by calculating the compensator argument at 2 kHz summed with the –179° of the power stage:

$$\arg G(f)=180+2\tan^{-1}\frac{2k}{300}-\tan^{-1}\frac{2k}{9.9k}-\tan^{-1}\frac{2k}{50k}-90\approx 239° \tag{9.92}$$

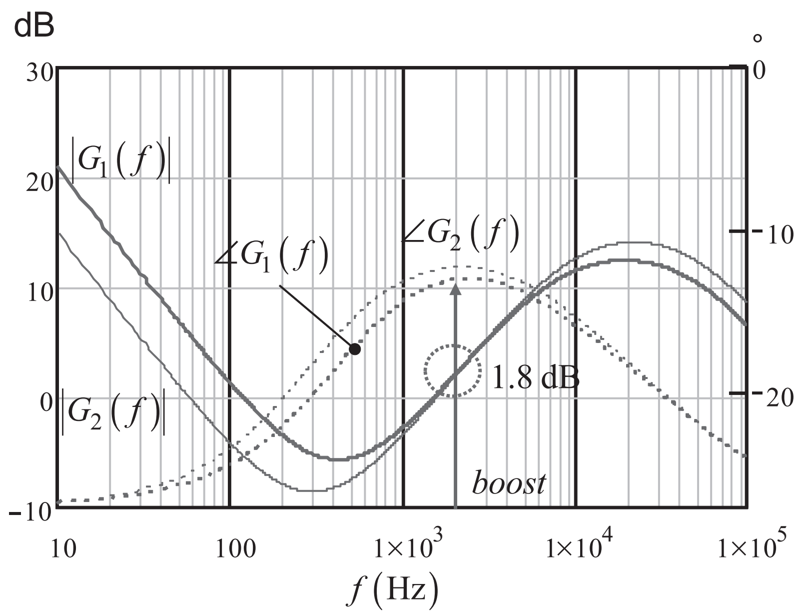

Figure 9.44 The type 3 compensator boosts the phase between the zeros and the poles. The first strategy offers a phase boost of 139°, whereas the second strategy increases it up to 149°, but to the detriment of the low frequency gain.

And check what the phase margin will be for a 11.5-V input voltage:

$$\varphi_m = \arg H(f_c) + \arg G(f_c) = 239 - 179 = 60° \quad (9.93)$$

If we compute it again at a 15-V input, we now have 66°, which is above the specification target. Both compensation options G_1 and G_2 appear in Figure 9.44.

If strategy 2 offers a more comfortable phase boost than strategy 1, please note the gain reduction in the low frequency region. This is the typical effect of moving the zeros farther down the frequency axis: you obtain a larger phase boost, but it will naturally affect the transient response by slowing down the system. Please note that both gain curves give exactly ≈1.8 dB at 2 kHz, compensating the gain deficiency of the power stage.

9.4.3 Plotting the Loop Gain

Once the compensator response is shaped according to your needs, you can now combine it with the boost converter frequency response and obtain the open-loop Bode plot we are looking for. It appears in Figure 9.45. Both strategies exhibit a crossover frequency exactly of 2 kHz, together with the phase margins we predicted.

Again, strategy 1 offers a larger low frequency gain but gives a phase margin below 60°. Strategy 2 improves the phase margin but slightly lowers the gain in the low frequency region.

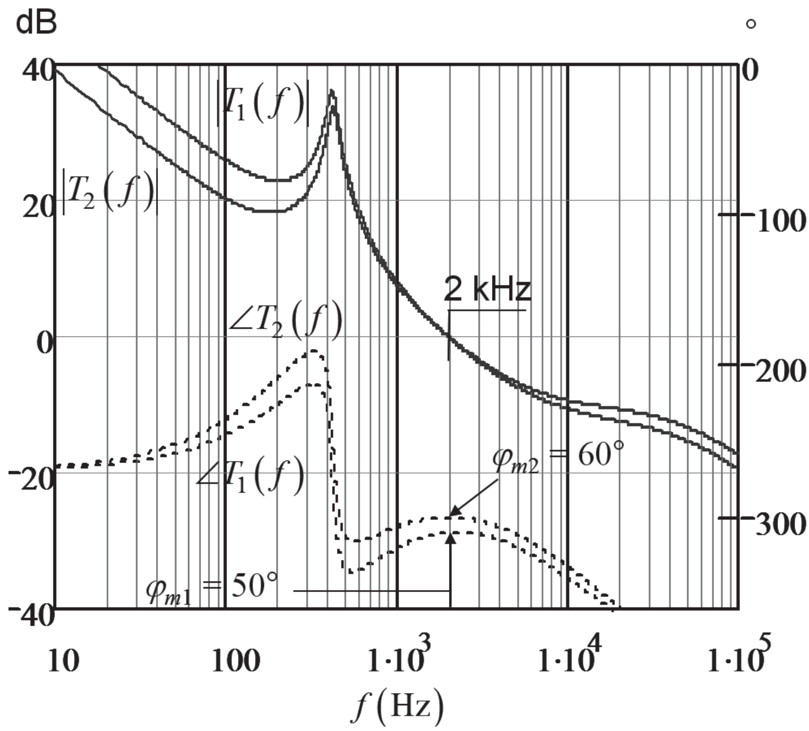

Figure 9.45 The loop gain combines $H(s)$ and $G(s)$ together. Strategy 1 and 2 give exactly a 2-kHz crossover frequency.

The calculations we have made are related to the typical ESR value of the output capacitor. As everyone knows, this ESR varies from lots to lots but also in temperature. The manufacturer gives you some indications on the minimum and the maximum of this ESR. It is important to perform the stability analysis with the ESR variations fully accounted for. In Table 9.1, we have assumed three different temperatures leading to three different ESR values. Then, thanks to the automated spreadsheet, we have entered these ESR values and collected the resulting phase margins depending on the two selected strategies.

As one can see, strategy 1 fails to offer the minimum acceptable phase margin at high temperature: we are below 45°. It does not say that the circuit will fail in production, but you start to have a design that can potentially be marginally stable. This kind of situation is not recommended for high-volume productions. On the contrary, strategy 2 offers enough margin, even at high temperature where we are still above 50°.

The complete study includes the operation of the boost converter when operated in light load conditions. Unfortunately, the converter will change its mode, transitioning from CCM to DCM. It can be shown that the transition occurs when the load reaches its critical point:

$$R_{\text{crit}} = \frac{2LF_{sw}}{D(1-D)^2} \tag{9.94}$$

It corresponds to a load of 69 Ω at a 11.5-V input voltage (5.2 W) or 76 Ω at a 15-V level (4.7 W). This is where the analytical analysis finds its limit: the transfer functions, both in ac and dc, change in DCM and all the equations must be updated. The new transfer function in DCM becomes:

$$\frac{V_{out}(s)}{V_{err}(s)} = H_0 \frac{\left(1+\frac{s}{\omega_{z_1}}\right)}{\left(1+\frac{s}{\omega_{p_1}}\right)} \tag{9.95}$$

where

$$H_0 = \frac{V_{in}}{V_{peak}} \frac{2}{2M-1} \sqrt{\frac{M(M-1)}{\tau_L}} \tag{9.96}$$

$$M = \frac{V_{out}}{V_{in}} \tag{9.97}$$

$$\tau_L = \frac{2L}{RT_{sw}} \tag{9.98}$$

Table 9.1 Phase Margin Variations Versus ESR Spreads

Temperature (°C)	*ESR* (mΩ)	*PM, strategy* **1** (°)	*PM, strategy* **2** (°)
0	40	62	72
25	20	50	60
70	10	43	53

$$\omega_{z_1} = \frac{1}{r_C C} \tag{9.99}$$

$$\omega_{p_1} = \frac{2M-1}{M-1}\frac{1}{RC} \tag{9.100}$$

You will have to take the compensator transfer function calculated to stabilized the CCM power stage and associate it with the transfer function of the power stage in DCM and check crossover and phase margin at both input voltage extremes. If the phase margin is acceptable, then you are done. If not, you will have to try different poles/zeros combinations so that stability is ensured regardless of the operating mode. This is not a Herculean labor, of course, but it clearly complicates and lengthens the stability analysis. This is where an auto-toggling SPICE model is of great help. Furthermore, it helps us predict the transient response in relationship to a set of operating conditions applied to the selected strategy.

9.5 Design Example 4: A Primary-Regulated Flyback Converter

There are ac-dc applications where using an optocoupler is not allowed. It can be for component cost reasons but also in cases where reliability is important, naturally banning optocoupler usage. For low-power applications, around 10W, the classical approach uses a primary-regulated flyback converter. Such a configuration appears in Figure 9.46. You can see two secondary windings: V_{out} is the main one, connected to the output load; it delivers the main power. V_{aux} is the auxiliary voltage supplying the controller and used for regulation purposes. As you can see, they do not share common ground for safety purposes. Isolation between the grounds is solely brought by the transformer. As indicated several times, modern PWM

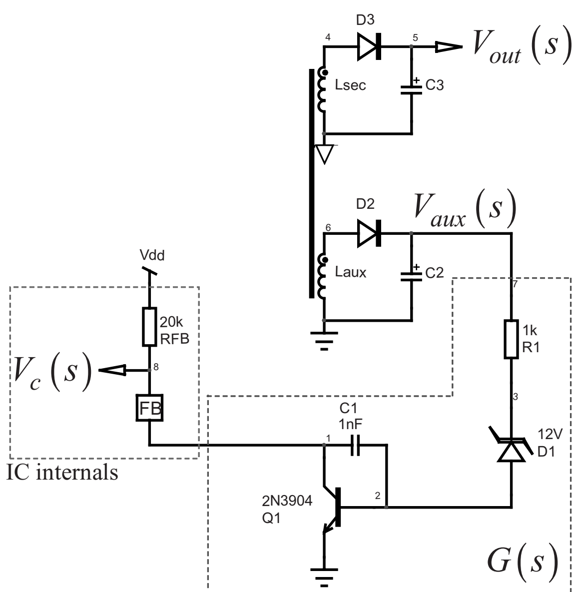

Figure 9.46 A primary-regulated flyback converter uses a simple transistor to regulate.

controllers do not embark op amp of any kind onboard. Capitalizing on the fact that all the regulation intelligence is kept on the secondary side, a simple feedback pin is enough to hook an optocoupler. Lacking this element, we will use a simple bipolar transistor to do pull-down job. Its base is driven by a Zener diode. It is a kind of low-gain shunt regulator, actually. The input of this error processing chain is the upper terminal of R_1, while the output is simply the control voltage V_c.

9.5.1 Deriving the Transfer Function

Most of the designers that I know have implemented this simple circuit by throwing component values based on their feelings! This is the wrong approach: you have to understand how the information is processed and see where weaknesses could hide. Otherwise, how can you stabilize a converter if you do not have a clue of the loop gain transfer function? Trial and error is not acceptable if you are a serious designer.

Based on this remark, we can draw a simplified small-signal schematic using the Ebers-Moll model for Q_1. The schematic is given in Figure 9.47.

At first glance, it looks like a piece of cake to solve such a transfer function. Well, we will see that fast analytical techniques will be of great help to carry the analysis. First, we count only one storage element, as this is a first-order system. Without knowing if it includes zeros or poles, it must obey the following transfer function:

$$G(s) = G_0 \frac{1 + s/\omega_z}{1 + s/\omega_p} = G_0 \frac{N(s)}{D(s)} \tag{9.101}$$

The first step in the process of deriving the transfer function is to get the dc gain G_0. This is done by opening all capacitors and shorting inductors if any. The new schematic becomes that of Figure 9.48.

The first equation to derive is the base current I_b:

$$I_b(s) = \frac{V_{aux}(s)}{r_d + R_1 + r_\pi} \tag{9.102}$$

Looking at the right side of the circuit, we can express the output voltage V_c:

$$V_c(s) = -\beta \cdot I_b(s) R_{FB} \tag{9.103}$$

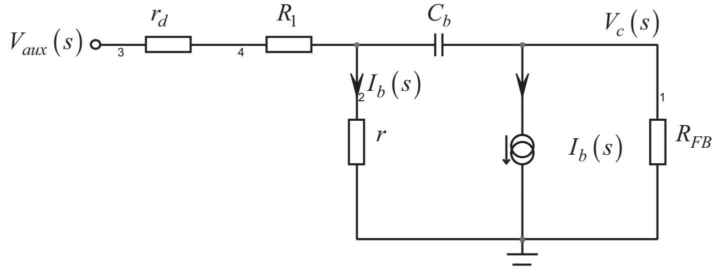

Figure 9.47 If we replace the transistor by its small-signal model, we obtain a simple sketch.

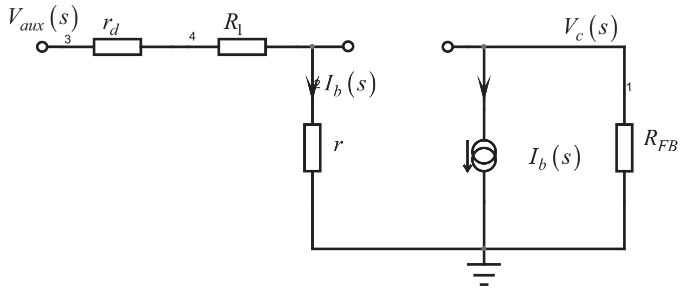

Figure 9.48 The dc transfer function is obtained by opening the capacitor.

Combining both equations, we now have for $s = 0$:

$$\frac{V_c(s)}{V_{aux}(s)} = G_0 = -\beta \cdot \frac{R_{FB}}{r_d + R_1 + r_\pi} \tag{9.104}$$

The capacitor is now put back in place and the schematic is slightly updated as shown in Figure 9.49. To check whether we have zero(s) in the transfer function, let's see if something in the signal path could prevent the excitation V_{aux} from reaching the output V_c.

If we have no response, it means that its upper terminal is 0 and no current flows through R_{FB}. The voltage across capacitor C_b is thus that across r_π:

$$V_\pi = r_\pi I_b(s) \tag{9.105}$$

$$V_{C_b} = \beta \cdot I_b(s) \frac{1}{sC_b} \tag{9.106}$$

As both voltages are equal, we have:

$$\beta \cdot I_b(s) \frac{1}{sC_b} = I_b(s) r_\pi \tag{9.107}$$

If we factor $I_b(s)$, we obtain the following expression:

$$I_b(s)\left[\beta \cdot \frac{1}{sC_b} - r_\pi\right] = 0 \tag{9.108}$$

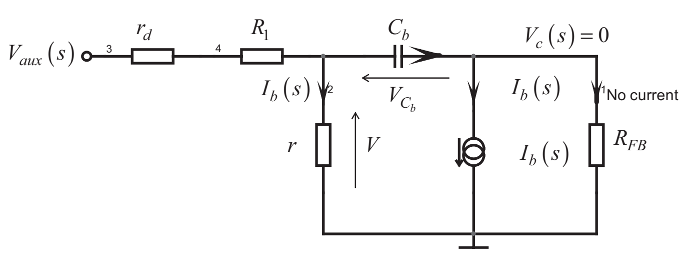

Figure 9.49 The exercise consists of finding what prevents the excitation from reaching the output.

which can only be realized if:

$$\beta \cdot \frac{1}{sC_b} - r_\pi = 0 \tag{9.109}$$

Solving for s gives us our zero location:

$$\omega_z = \frac{\beta}{C_b h_{11}} \tag{9.110}$$

This is a positive root; we have a nice RHPZ. The numerator can thus be put under the following form:

$$N(s) = 1 - s/\omega_z \tag{9.111}$$

Now that we have our numerator, we can look at the denominator $D(s)$. The denominator of a network depends only on its structure, not on the excitation signal. Regardless of whether you excite it at different inputs, let's say to unveil an output impedance, an input admittance, and so on, the denominator of the obtained transfer function will remain unchanged. The excitation signal can therefore be put to zero: open a current source and short circuit a voltage source. This is what is shown in Figure 9.50, further updated in Figure 9.51 where we lumped r_d and R_1 into a single resistor R:

$$R = r_d + R_1 \tag{9.112}$$

To get the denominator expression, we need to find the time constant in which C_b is associated with the equivalent resistor R_e driving it. In other words, let us derive the resistor expression seen at the capacitor terminals when we removed it from the picture. To derive an impedance, we excite its terminals by a current source (the "excitation") and we observe the voltage developed across the current source terminals: the "response." In Figure 9.51, we respectively called these signals $I_T(s)$ and $V_T(s)$.

Let's write some basic equations, starting with the voltage across the resistor R, on the left of the figure:

$$V_R(s) = (I_T(s) + I_b(s))R \tag{9.113}$$

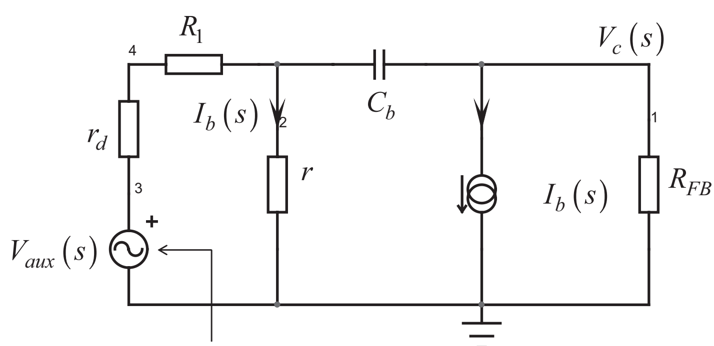

Figure 9.50 The denominator is obtained by setting the excitation signal to zero.

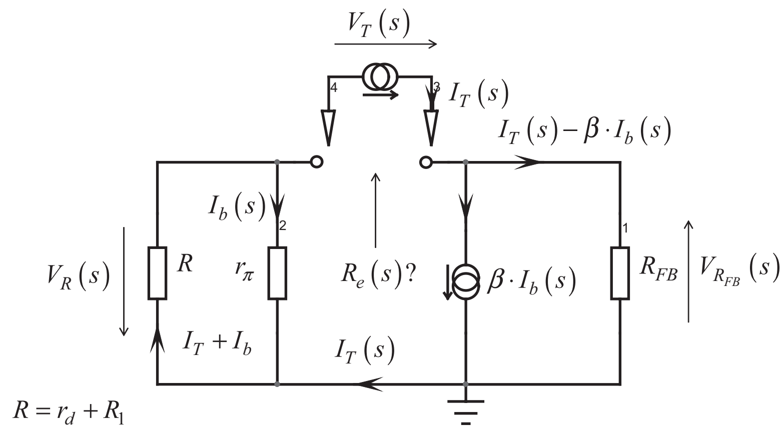

Figure 9.51 The excitation is now set to zero, and the lower terminal of r_d has been grounded.

This voltage is exactly the same that you will find across r_π but with an opposite sign:

$$V_R(s) = -r_\pi I_b(s) \tag{9.114}$$

Equating these two equations and solving for the have:

$$I_b(s) = -\frac{R \cdot I_T(s)}{R + r_\pi} \tag{9.115}$$

The voltage across the pull-up resistor RFB comes easily:

$$V_{R_{FB}}(s) = (I_T(s) - \beta \cdot I_b(s))R_{FB} \tag{9.116}$$

The voltage across the injection source V_T is not difficult either:

$$V_T(s) = V_{R_{FB}}(s) + V_R(s) \tag{9.117}$$

Now substituting (9.116), (9.114), and (9.115) into (9.117), we obtain:

$$V_T(s) = (I_T(s) - \beta \cdot I_b(s))R_{FB} - r_\pi I_b(s) \tag{9.118}$$

$$V_T(s) = \left(I_T(s) + \beta \frac{R \cdot I_T(s)}{R + r_\pi}\right)R_{FB} + r_\pi \frac{R \cdot I_T(s)}{R + r_\pi} = I_T(s)\left(\left(1 + \frac{\beta \cdot R}{R + r_\pi}\right)R_{FB} + \frac{R \cdot r_\pi}{R + r_\pi}\right) \tag{9.119}$$

Rearranging this equation, we have the port input impedance:

$$\frac{V_T(s)}{I_T(s)} = \left(1 + \beta \frac{R}{R + r_\pi}\right)R_{FB} + R \parallel r_\pi \tag{9.120}$$

The denominator can now be written under the following form:

$$D(s) = 1 + s/\omega_p \tag{9.121}$$

in which the pole is:

$$\omega_p = \frac{1}{C_b\left[\left(1+\beta\frac{r_d+R_1}{r_d+R_1+r_\pi}\right)R_{FB}+\left(r_d+R_1\right)\|\, r_\pi\right]} \tag{9.122}$$

The complete transfer function for Figure 9.47 is thus:

$$\frac{V_{FB}(s)}{V_{aux}(s)} = -\beta\cdot\frac{R_{FB}}{r_d+R_1+r_\pi}\frac{1-\dfrac{s}{\dfrac{\beta}{C_b h_{11}}}}{1+sC_b\left[\left(1+\beta\dfrac{r_d+R_1}{r_d+R_1+r_\pi}\right)R_{FB}+\left(r_d+R_1\right)\|\, r_\pi\right]} \tag{9.123}$$

This equation appears in a clear and ordered way, where you can see the pole and zero locations. This is the strength of the fast analytical techniques.

9.5.2 Verifying the Equations

When running this type of analysis, it is always possible to overlook a condition or a parameter, or simply make a mistake when developing equations. For all of these reasons, I will verify the ac response of (9.123) and confront it to a simple SPICE simulation. If all is correct, the curve must perfectly superimpose.

The Mathcad® sheet appears in Figure 9.52. With the values entered in the automated sheet, the RHPZ shows up at around 550 kHz. It is obviously of negligible influence if we plan to crossover below 10 kHz. We adopted arbitrary values for all components.

$\beta := 73 \quad h_{11} := 21.4\text{k}\Omega \quad R_1 := 1\text{k}\Omega \quad r_d := 12.6\text{k}\Omega \quad C_b := 1\text{nF} \quad R_{FB} := 20\text{k}\Omega$

$$G_0 := \frac{\beta\cdot R_{FB}}{h_{11}+r_d+R_1} = 41.714 \qquad 20\cdot\log\left(G_0\right) = 32.406$$

$$f_z := \frac{\beta}{C_b\cdot h_{11}\cdot 2\pi} = 542.912\text{kHz}$$

$$f_p := \frac{1}{2\cdot\pi\cdot C_b\cdot\left[\left(1+\beta\cdot\dfrac{r_d+R_1}{r_d+R_1+h_{11}}\right)\cdot R_{FB}+\dfrac{\left(r_d+R_1\right)\cdot h_{11}}{r_d+R_1+h_{11}}\right]} = 267.205\text{Hz}$$

$$G_1(s) := -G_0\cdot\frac{1-\dfrac{s}{2\cdot\pi\cdot f_z}}{1+\dfrac{s}{2\cdot\pi\cdot f_p}}$$

Figure 9.52 The Mathcad® sheet shows the presence of the pole and the RHPZ.

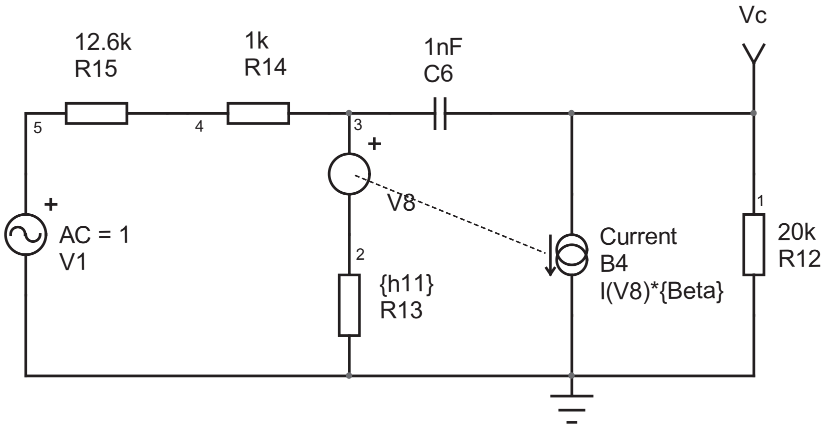

Figure 9.53 The SPICE simulation template is quickly captured.

The SPICE simulation of the Figure 9.47 circuit is not complicated, as shown in Figure 9.53. The simulation time is flashing; there is no switching component. We can now compare the Mathcad® results with those given by SPICE, and we can see in Figure 9.54 that the curves are similar leading to the conclusions that our equations are correct. It does not mean that the theory is correct! The real referee is the bench measurement that needs to agree with the theoretical results. What we have done here is just a simple analytic check that you need to do anyway.

9.5.3 Stabilizing the Converter

We have our compensator transfer function with (9.123). It is now time to tailor it in order to reach our crossover goal, together with an adequate phase margin. Needless to say, a simple bipolar transistor together with a Zener diode won't give us the same performance as with an op amp, but let's see what we can do anyway.

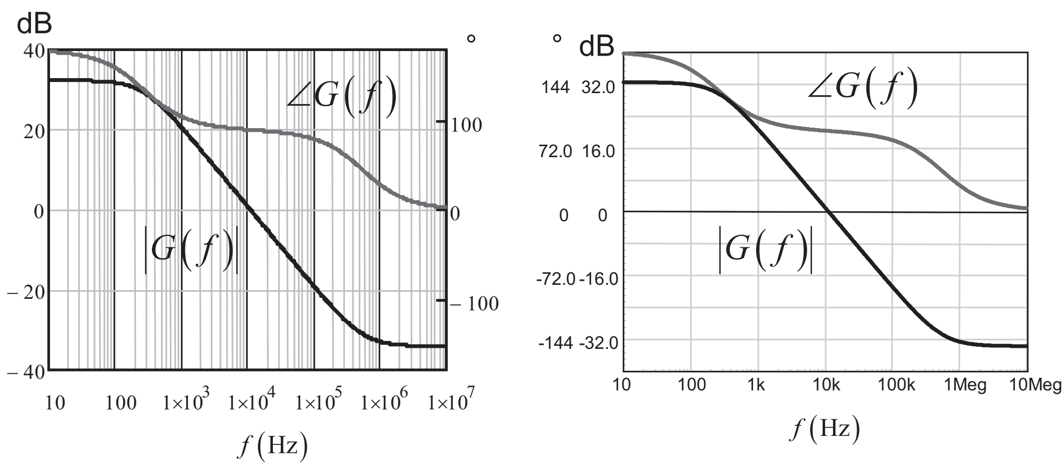

Figure 9.54 The results delivered by Mathcad® and SPICE are identical; the derived equations are correct.

First, in a primary-regulated converter, the winding on which you monitor the output voltage image is the auxiliary winding: you do not see directly the effects of the load on the output voltage, but an image translated by the turns ratio between the power and auxiliary windings. To perform small-signal analysis, you must account for this relationship and perform a full reflection of what is installed on the output to the auxiliary side. The implementation of such a reflection process appears in Figure 9.55. You have to individually reflect the power elements (the load and the filtering capacitor) to the winding where regulation is performed. As this auxiliary winding is already equipped with a capacitor and an equivalent resistor (the controller consumption for instance), the reflected elements combine with those already in place. Please note that the ESR and capacitors combinations shown in the schematic work only if the time constants $r_{C_1}C_1$ and $r_{C_2}C_2$ are equal. If they are not, it is an approximation. Furthermore, when reflecting the elements, we consider the diode dynamic resistance small enough to neglect it. This is not always the case, in particular in light-load operation.

Once you have reduced the current-mode flyback converter to a single winding type, you can use the small-signal CCM current-mode power stage transfer equation such as that given in [12]:

$$H(s) = H_0 \frac{\left(1+\frac{s}{\omega_{z1}}\right)\left(1-\frac{s}{\omega_{z2}}\right)}{1+\frac{s}{\omega_{p1}}} \tag{9.124}$$

$$H_0 = \frac{R_{eq}}{R_{sense}G_{FB}N} \frac{1}{\frac{(1-D)^2}{\tau_L}+2M+1} \tag{9.125}$$

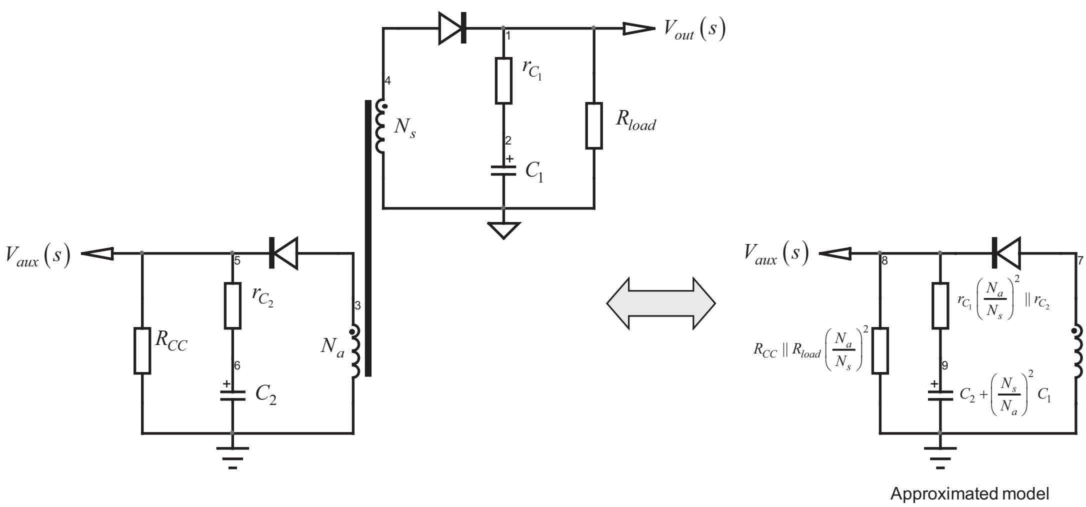

Figure 9.55 Reflection of the output elements across the auxiliary windings is necessary to perform the small-signal analysis.

$$f_{z_2} = \frac{(1-D)^2 R_{eq}}{2\pi D L_p \left(\frac{N_a}{N_p}\right)^2} \tag{9.126}$$

$$f_{p_1} = \frac{\frac{(1-D)^3}{\tau_L} + 1 + D}{2\pi R_{eq} C_{eq}} \tag{9.127}$$

$$f_{z_1} = \frac{1}{2\pi r_{C_{eq}} C_{eq}} \tag{9.128}$$

in which:

$$r_{C_{eq}} = r_{C_1}\left(\frac{N_a}{N_s}\right)^2 \parallel r_{C_2} \tag{9.129}$$

$$C_{eq} = C_2 + C_1\left(\frac{N_s}{N_a}\right)^2 \tag{9.130}$$

$$R_{eq} = R_{CC} \parallel R_{load}\left(\frac{N_a}{N_s}\right)^2 \tag{9.131}$$

$$\tau_L = \frac{L_p\left(\frac{N_a}{N_p}\right)^2}{R_{eq} T_{sw}} \tag{9.132}$$

You can clearly imagine how tedious it quickly becomes to calculate these contributions as you move some parameters to see their effects on stability. Even with an automated sheet, it is not that friendly. Furthermore, if the converter transitions to DCM, the equation set is no longer valid and must be updated.

The most convenient solution is to use SPICE with an auto-toggling DCM/CCM model derived for current mode (see [12]). The simulation template appears in Figure 9.56.

We have picked a standard NPN transistor to which a 12-V Zener diode has been added. The multiwinding transformer is built by adding a simple transformer model in parallel with the existing one. All the elements found on the power winding are automatically reflected on the auxiliary winding during simulation. Even better, the reflection includes the dynamic resistor of the rectifying diode that can easily change depending on the operating current: from an almost short circuit at high current, it can become a larger value in light load conditions, and our reflection formulas no longer work. On the contrary, SPICE will account for this change in the series resistance, leading to a correct power stage ac transfer function at all possible operating points. The power stage transfer function is unveiled in Figure 9.57. Please note that we probed node 10 (on the auxiliary winding) to graph the transfer function.

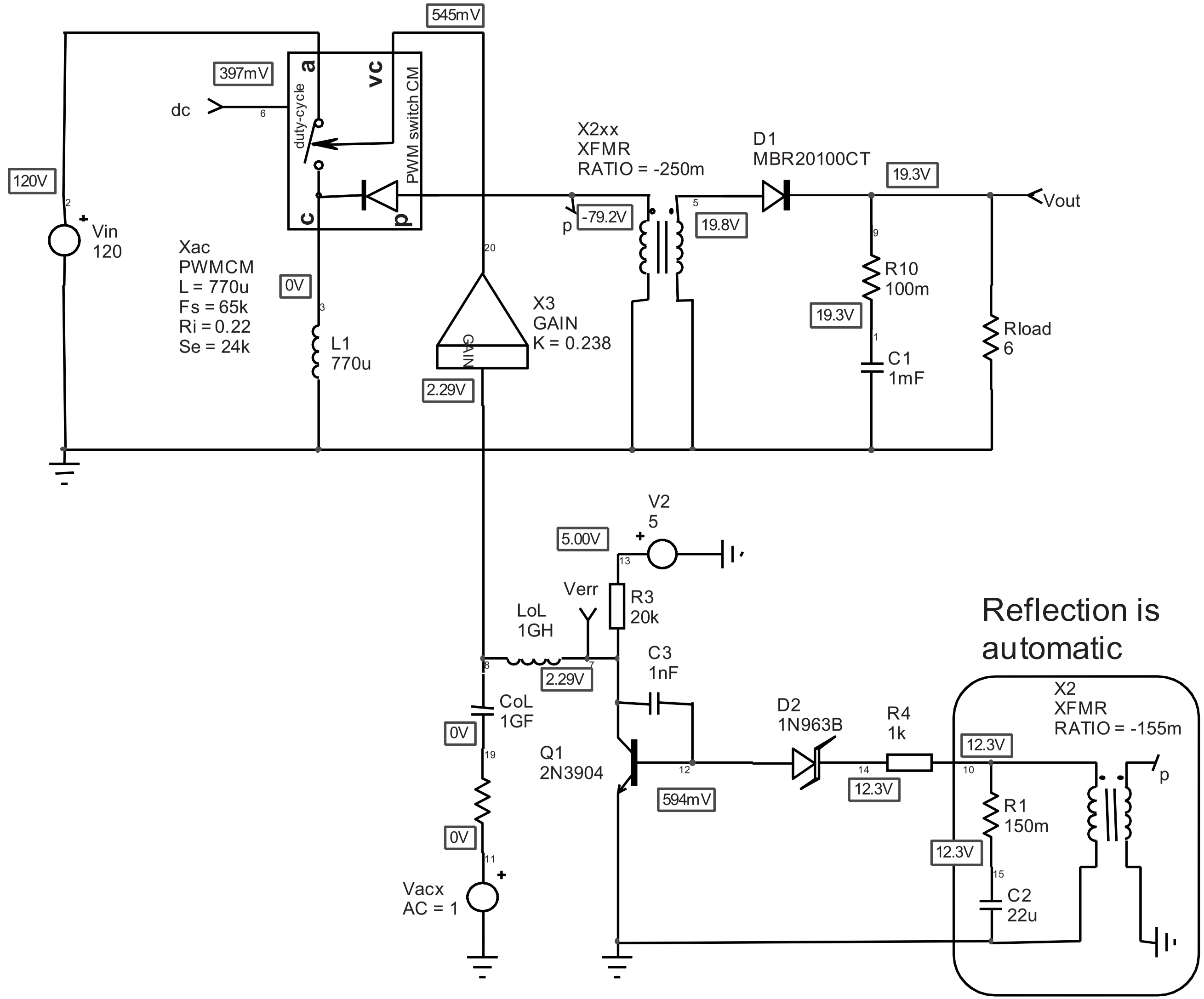

Figure 9.56 SPICE does all the reflections for you, and parameters sweep is a child's play.

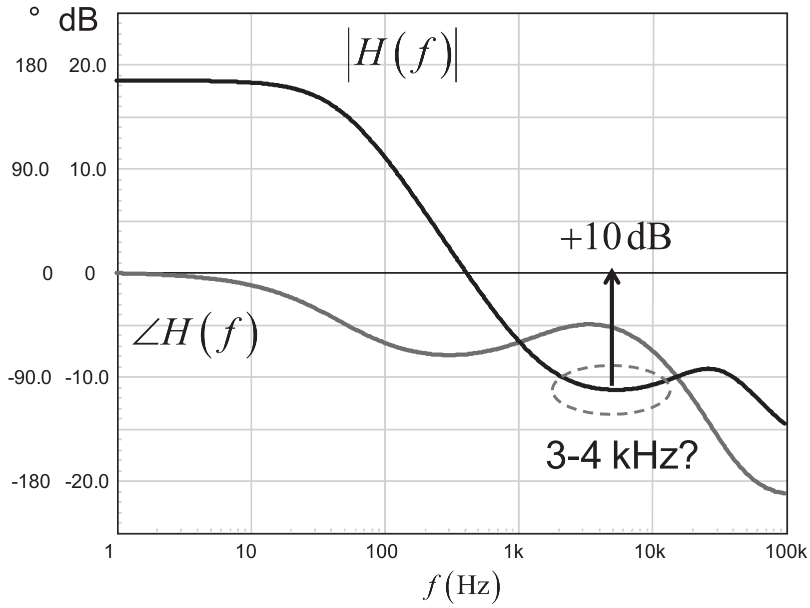

Figure 9.57 The power stage shows a first-order response with a minimum phase lag around 3–4 kHz.

As the adopted compensator cannot provide a phase boost of any kind, we will select a crossover area at a position where the phase lag is not too strong. Around 3–4 kHz seems to be the right region. If we neglect the high-frequency RHPZ contribution, we have a single-pole response as confirmed in Figure 9.54. We will therefore place the pole with C_b at a position where the gain provided by the compensator matches the 10-dB gain deficit indicated in Figure 9.57.

If we neglect the RHPZ contribution, the compensator transfer function described by (9.101) can be simplified to:

$$G(s) \approx G_0 \frac{1}{1 + s/\omega_p} \tag{9.133}$$

The magnitude at crossover is found by replacing s by $j\omega_c = j2\pi f_c$:

$$|G(f_c)| \approx G_0 \frac{1}{\sqrt{1 + \left(\frac{f_c}{f_p}\right)^2}} \tag{9.134}$$

From this value, we can extract the position of the pole based on the needed gain at f_c:

$$f_p = \frac{f_c \cdot |G(f_c)|}{\sqrt{G_0{}^2 - |G(f_c)|^2}} \tag{9.135}$$

Now, assume the following components values:

$R_1 = 1\ \text{k}\Omega$; Zener series resistor

$R_{FB} = 20\ \text{k}\Omega$; pull-up resistor

$r_d \approx 10\ \text{k}\Omega$; Zener diode dynamic resistor

$\beta \approx 70$; transistor current gain

$r_\pi \approx 21\ \text{k}\Omega$; transistor dynamic base resistance

We can calculate the compensator plateau gain G_0:

$$G_0 = \beta \cdot \frac{R_{FB}}{r_d + R_1 + r_\pi} = 73 \times \frac{20k}{10k + 1k + 21k} = 45.6 = 33\text{dB} \tag{9.136}$$

According to (9.135), the pole will have to be placed at

$$f_p = \frac{3.3k \times 3.16}{\sqrt{45.6^2 - 3.16^2}} \approx 230\text{Hz} \tag{9.137}$$

To calculate the capacitor value, we will use (9.122) from which we calculate the equivalent resistor value driving the capacitor C_b:

$$R_{eq} = \left(1 + \beta \frac{r_d + R_1}{r_d + R_1 + r_\pi}\right) R_{FB} + \left(r_d + R_1\right) \| \, r_\pi \approx 595\ \text{k}\Omega \tag{9.138}$$

The capacitor value is thus:

$$C_b = \frac{1}{2\pi f_p R_{eq}} = \frac{1}{6.28 \times 230 \times 595k} \approx 1.2n\text{F} \tag{9.139}$$

If we set this component values in our simulation template of Figure 9.56 and check the collector signal, we will have a view of the compensated loop gain. It appears in Figure 9.58.

It is not exactly the crossover frequency we wanted, but given the dispersion on the transistor gain and the various dynamic resistors, it is not that bad! A final step load simulation shows that the transient response is acceptable for such a simple system. The resulting waveform is in Figure 9.59. Please note that the average model does not include the various transformer leakage inductances that affect the transient response.

Some people will want to increase the Zener current, forcing it to work farther off its knee. In this case, simply install a resistor R_B across the transistor base-emitter. It will impose a Zener current at room temperature of:

$$I_Z \approx \frac{0.65}{R_B} \tag{9.140}$$

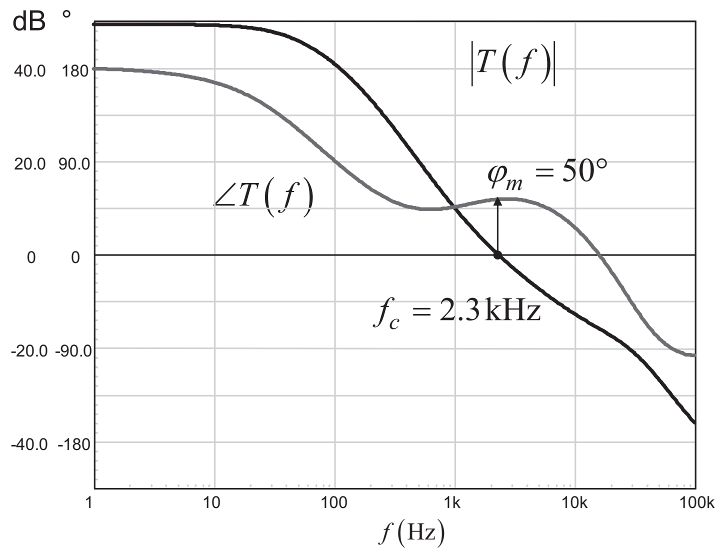

Figure 9.58 The compensated loop gain shows a good phase margin with a decent crossover frequency.

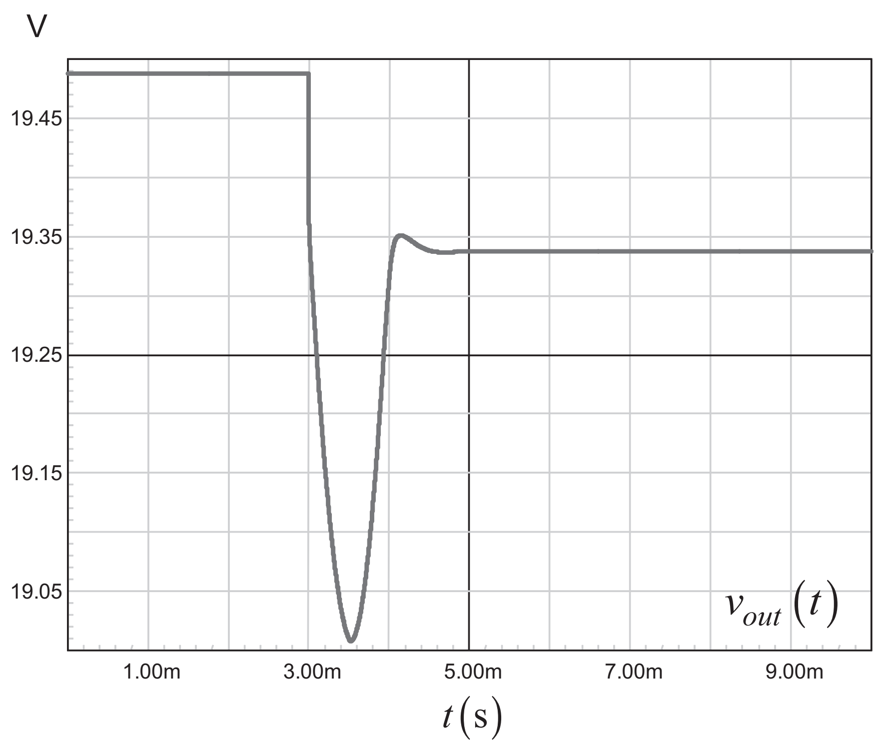

Figure 9.59 The transient response is acceptable for this primary-regulated converter. The step load is 1.5 A in 10 μs.

The transfer function is slightly modified by this resistor addition. If you go trough the equations again or look at page 53 of [15], you should find:

$$G_0 = \frac{1}{1+\dfrac{R}{R_B}} \frac{\beta R_{FB}}{r_\pi \left(1+\dfrac{R \,||\, R_B}{r_\pi}\right)} \tag{9.141}$$

Its limit when R_B goes infinite is given by (9.104). The RHPZ position remains unchanged and (9.110) is still valid. The pole position is, however, slightly affected:

$$\omega_p = \frac{1}{C_b\left(R_{FB}+\left(1+\dfrac{\beta R_{FB}}{r_\pi}\right)\cdot r_\pi \,||\, (R \,||\, R_B)\right)} \tag{9.142}$$

In these expressions, $R = r_d + R_1$.

9.6 Design Example 5: Input Filter Compensation

Switching converters are noisy systems in essence. The sharp voltage or current interruptions generate differential or common mode noise that can couple to the supply wires in certain conditions. If low-noise systems share a common supply line with a switching converter, it is very likely that the switching converter pollutes the common supply rail and alters the sensitive equipments behaviors. To avoid high-frequency pulses to flow back to the dc source and pollute the supply rail, an input filter has to be installed. A typical filter appears in Figure 9.60. To minimize losses, it is usually made of an LC network. This network forms a second-order filter and prevents the noise from returning to the generator as ac currents are supposed to circulate in C while L opposes a high impedance to these currents.

Calculating the cutoff frequency of such a filter depends on the maximum input ripple specifications and is outside the scope of this book. Reference [12] presents several design examples based on a specification to fulfill. What we will see now is the problem brought by the filter insertion and how to cure it.

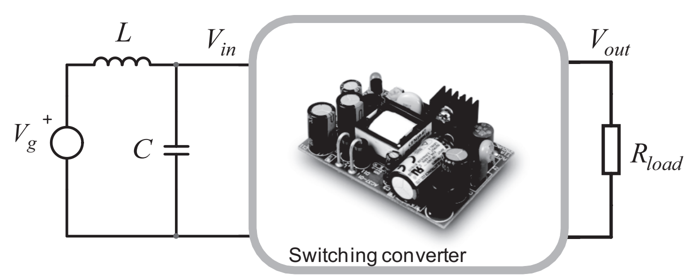

Figure 9.60 A typical filter is made by combining L and C elements.

9.6.1 A Negative Incremental Resistance

A switching converter is a closed-loop system. It makes sure that the output variable (e.g., V_{out}) stays in control regardless of incoming perturbations such as the input voltage V_{in} or the output current I_{out}: if you deliver a certain amount of power, whether it is low line or high line, the converter will feed the load at constant power. If we have 100 percent efficiency, we can write:

$$P_{in} = P_{out} = I_{in}V_{in} = I_{out}V_{out} \tag{9.143}$$

The input current can thus be derived as

$$I_{in}(V_{in}) = \frac{P_{out}}{V_{in}} \tag{9.144}$$

If we plot the input current versus the input voltage, we obtain the curve shown in Figure 9.61. If the input voltage increases as the output power is constant (10 W), then the input current must go down to satisfy (9.143). On the opposite, if the input voltage decreases, then the loop forces the current to increase. If we are interested in the slope of this curve, we must derive (9.144):

$$\frac{dI_{in}(V_{in})}{dV_{in}} = \frac{d\left(\dfrac{P_{out}}{V_{in}}\right)}{dV_{in}} = -\frac{P_{out}}{{V_{in}}^2} \tag{9.145}$$

This is the negative relationship that links the current to the voltage when the input line changes. The slope unit is amperes per volts, or a conductance. If we take the inverse of it, we have an expression that we will call the *incremental input resistance*.

Its expression is simply:

$$R_{in,inc} = -\frac{{V_{in}}^2}{P_{out}} \tag{9.146}$$

The concept requires a bit of abstraction. If the converter would offer a truly resistive input that absorbs power, then the current would exclusively depend on the input voltage. With a closed-loop converter, the input current depends on the input voltage, of course, but also on the constant output power controlled by the

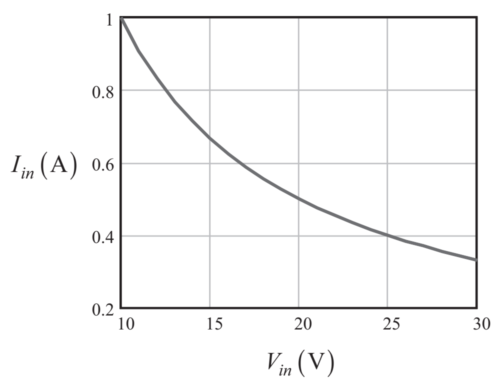

Figure 9.61 The relationship between the input current and the input voltage shows a negative slope.

system. The negative incremental resistance value expressed in (9.146) is the result of this process.

9.6.2 Building an Oscillator

An oscillator can be formed in different ways. One is to use a tuned LC network. If you bring an initial energy to the network with a fugitive stimulus, oscillations take place under the form of energy that swings back and forth between the storage elements L and C. Oscillations cease when the energy stored at the beginning of the oscillations have been totally dissipated in the ohmic paths such as capacitor and inductor ESRs. If you do not want the oscillations to cease, the energy transfer between L and C must be lossless. In other words, if you compensate the losses by canceling the circuit resistances, oscillations once started will be continuously maintained.

One way to cancel the ohmic path is to associate a negative resistor to the circuit. A negative resistor can be emulated using semiconductors (FETs, Gunn and tunnel-effect diodes, negative impedance converter op amp-based circuits, and so on). As an example, Figure 9.62 portrays an LC network where the losses are introduced by the loading resistor R. The quality factor in this example is adjusted to 5. The network is briefly excited by a 5-V step of a few μs duration. Figure 9.63 shows how the oscillations take place and die out after several cycles, indicative of energy dissipated in the resistor.

The graph shows you how the energy transfers back and forth between the capacitor and the inductor. The total energy stored in the circuit at any time (w_{tot}) is the sum of the energy individually stored in L (w_L) and C (w_C):

$$w_{tot}(t) = w_L(t) + w_C(t) = \frac{1}{2}L{i_L}^2(t) + \frac{1}{2}C{v_C}^2(t) \tag{9.147}$$

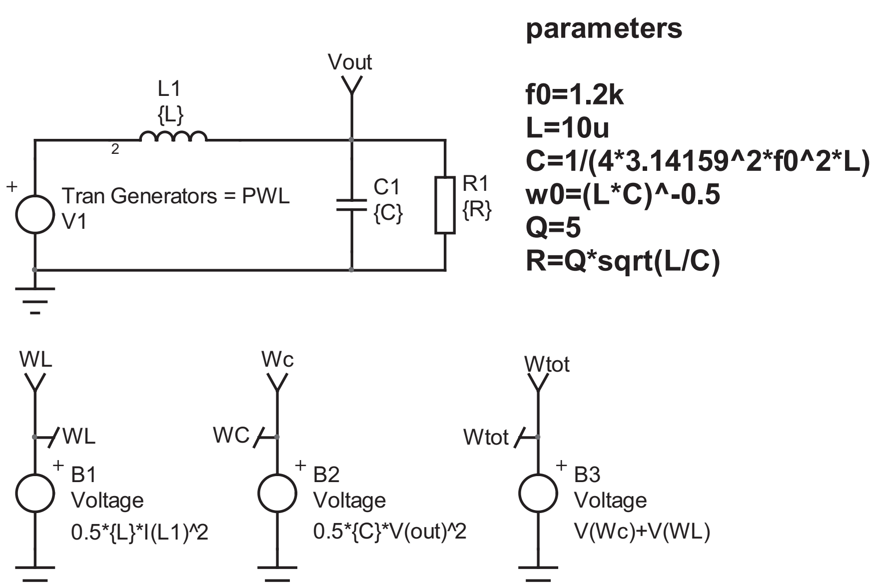

Figure 9.62 In this circuit, the losses damp the network and oscillations quickly die out.

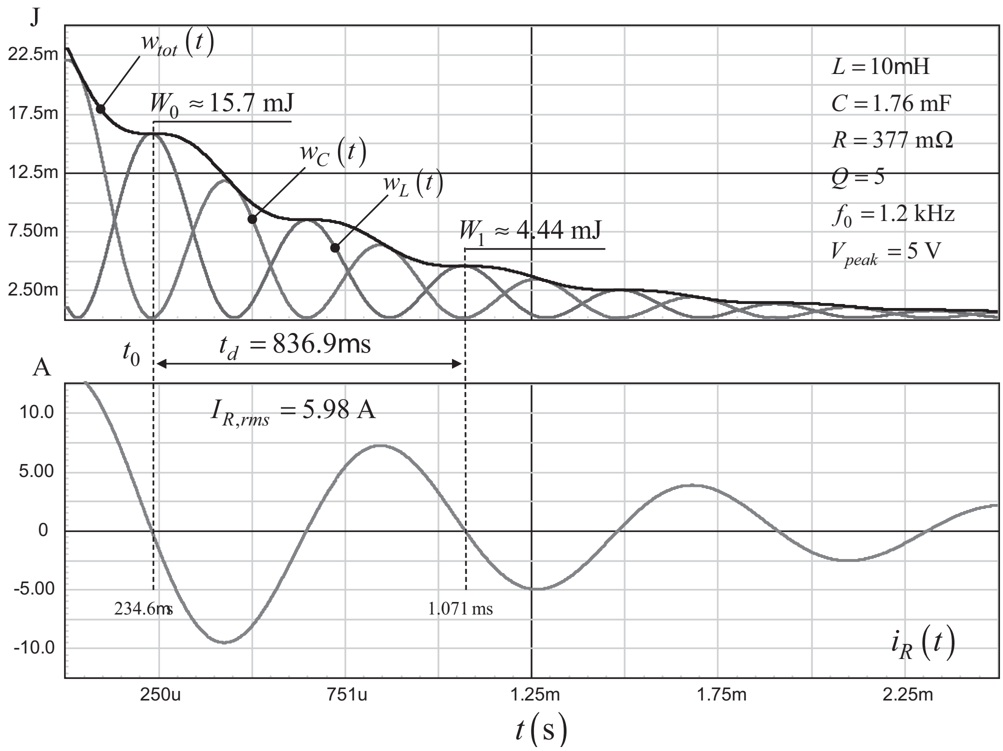

Figure 9.63 Because of the losses incurred to the resistor, the energy exchange between the inductor and the capacitor is affected and oscillations cease quickly.

As the series resistor R_1 dissipates the losses in heat, the amount of stored energy drops as time elapses. If we look at the figure, the energy lost between t_0 and $t_0 + t_d$ is:

$$\Delta W = w(t_0) - w(t_0 + t_d) = 15.7m - 4.44m = 11.26 \text{ mJ} \tag{9.148}$$

The power dissipated in the resistor during a complete pseudo period is simply:

$$P_{R_1} = R_1 {I_{L,rms}}^2 = 0.377 \times 5.98^2 = 13.467 \text{W} \tag{9.149}$$

The corresponding energy lost in the resistor is given by:

$$W_{R_1} = P_{R_1} t_d = 13.48 \times 836.9u = 11.27 \text{ mJ} \tag{9.150}$$

This corresponds to what was found in (9.148).

From this graph, we can derive the quality factor. The quality factor Q is a parameter that quantifies the reactive component in the considered circuit. If high, reactive phenomena dominate over resistive losses. If low, the circuit is weakly reactive and losses are important. Q can be defined as follows:

$$Q = 2\pi \frac{\text{(stored energy)}}{\text{(energy dissipated per cycle)}} \tag{9.151}$$

In our circuit under a transient excitation, the considered energy is that stored in the capacitor and the inductor over the time t_d. By placing the cursors on the w_{tot} curve of Figure 9.63 and computing the average value over t_d, we read:

$$\langle w_{tot}(t)\rangle_{t_d} = 9.09 \text{ mJ} \tag{9.152}$$

If we apply (9.151), we have:

$$Q = 2\pi\frac{\text{(stored energy)}}{\text{(energy dissipated per cycle)}} = 6.28\times\frac{9.09m}{11.27m} = 5.06 \tag{9.153}$$

The quality factor could also be obtained using the logarithmic decrement δ derived in Chapter 3. The added 2 in the numerator accounts for the energy terms rather than currents or voltages:

$$Q = \sqrt{\left(\frac{2\pi}{\ln\left(\frac{W_0}{W_1}\right)}\right)^2 + 0.25} = \sqrt{\left(\frac{2\,\pi}{\ln\left(\frac{15.715m}{4.444m}\right)}\right)^2 + 0.25} = 5 \tag{9.154}$$

Analytically, we can show that the quality factor of the RLC circuit from Figure 9.62 under harmonic excitation is:

$$Q = R\sqrt{\frac{L}{C}} \tag{9.155}$$

If we place in parallel with R a resistor value $-R$, the quality factor becomes

$$Q = \frac{-R^2}{R-R}\sqrt{\frac{L}{C}} = \infty \tag{9.156}$$

We have updated the simulation circuit from Figure 9.62 where a resistor of value –377 mΩ has been installed across R_1. The new simulation results appear in Figure 9.64 and confirm the presence of sustained oscillations. The current diverted by the paralleled combination of R and $-R$ is zero. Should we derive the roots expression of the new RLC circuit featuring the added resistor of $-R$ value, we would find imaginary poles, with nullified real parts. We have created a negative resistance oscillator.

9.6.3 Taming the Oscillations

In the previous example, a (positive) resistor R_1 was already in place across the output capacitor, efficiently damping the network (Q = 5). What if the negative resistor, alone, loads the LC filter? This is actually the Figure 9.60 illustration with a closed-loop converter loading an undamped filter. If you run the analysis of an LC filter loaded by a negative resistor, you will find that the poles are no longer pure imaginary values but have jumped on the right side of the vertical axis: they become right half plane poles or positive roots. If you remember the transient response of a system featuring RHP poles, the exponent of the exponential term(s) is positive,

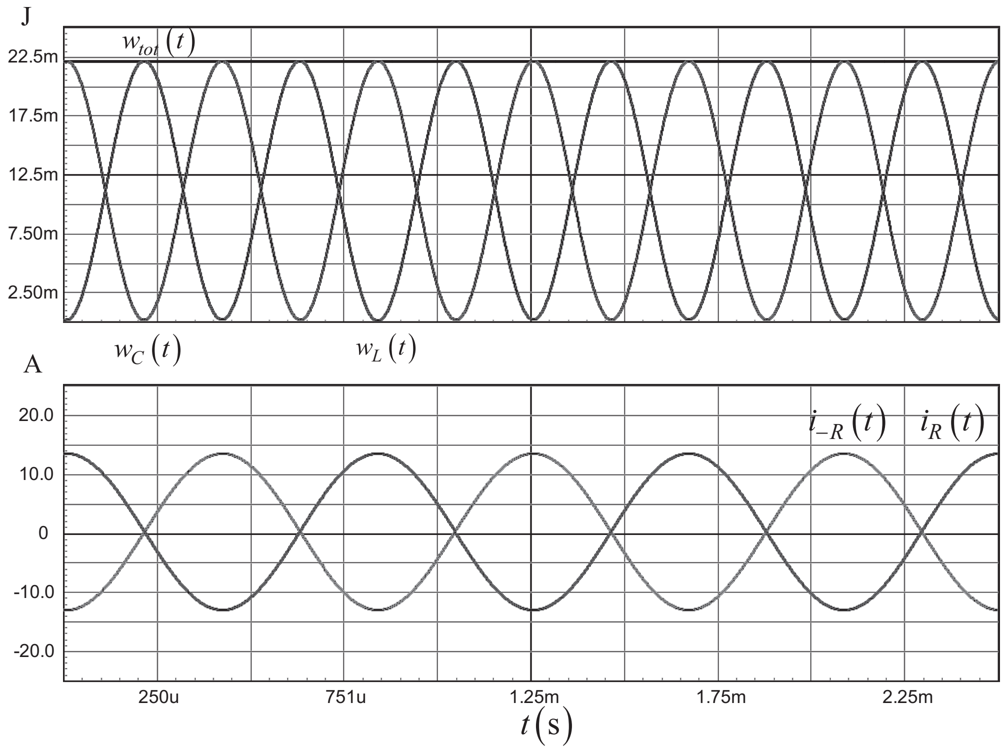

Figure 9.64 The negative resistance actually cancels the current diverted from the capacitor, bringing down the losses to 0 W.

meaning the system response diverges with time. This is what is confirmed by Figure 9.65. In a real system, the physical limits (i.e., power supply voltage) will clamp the excursion and the converter is likely to lock out. In high-power converters, the limit might be far away from the semiconductor maximum ratings: you should expect a loud noise followed by smoke.

To lower the quality factor, (9.151) tells us that we must create losses in the filter. These losses will dissipate some stored energy in heat, and oscillations will cease. Many solutions are possible, and [16] explores several circuits. The key is to maintain efficiency despite damping effects. One easy way to create a loss is to add a damping resistor across the inductor. As the inductor voltage at steady-state is zero, losses will only occur in presence of oscillations. Unfortunately, associating the damping resistor with the inductor introduces a zero in the transfer function counteracting one of the LC network double poles: from a second-order network, the filter becomes a first-order network, clearly affecting its filtering capabilities.

Another option consists of adding the damping resistor in parallel with the output capacitor. As ac damping is needed (when oscillations appear), you can insert this resistor in series with it a dc-block capacitor. Therefore, in dc, the resistor will be transparent to the circuit dissipating no power. Losses will occur only if ac oscillations arise. The updated filter sketch is given in Figure 9.66.

The calculation of the damping resistor is rather simple. Let's assume we have a 5-V/30-A ac-dc converter running from US mains. The lowest input voltage is 100 V

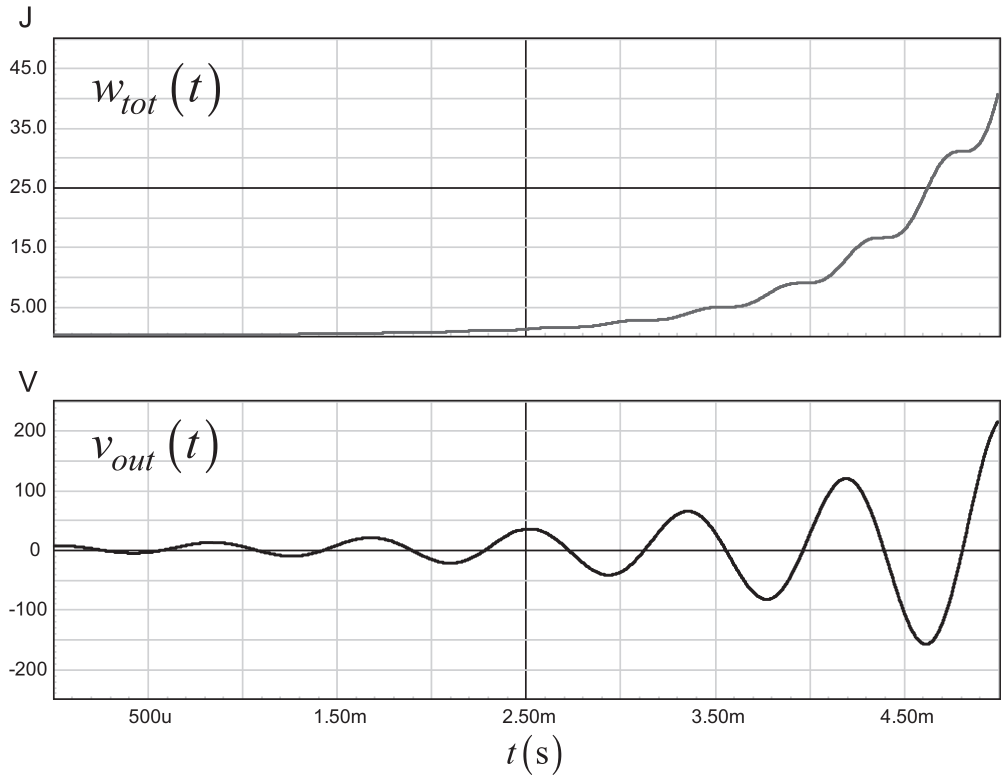

Figure 9.65 When the negative resistor alone loads the *LC* network, the output voltage diverges as *t* increases.

rms, or approximately a 100-V rectified dc rail, if we consider a 30 percent ripple on the bulk capacitor. Considering a 95 percent efficiency, the incremental input resistance of this converter is:

$$R_{in} = -\frac{V_{in}{}^2}{P_{out}}\eta = -\frac{100^2}{150}0.95 \approx -63\ \Omega \tag{9.157}$$

The *LC* network selected for this filter is a 150-μH inductor coupled to a 10-μF multilayer capacitor. We assume there is no ESR associated with these elements; this

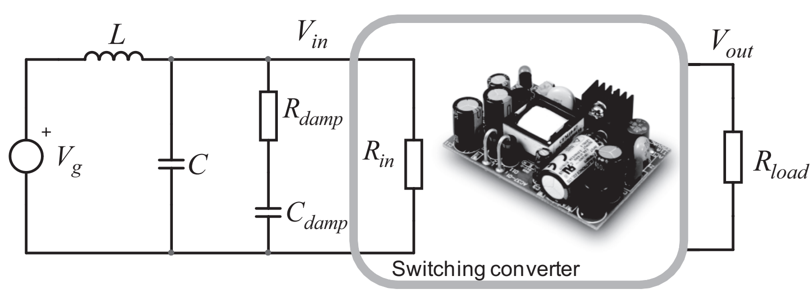

Figure 9.66 The resistor is dc-isolated from the supply rail and only couples in ac. Efficiency is less affected than with a series resistor.

is the worst case. If we use (9.155), we can update the quality factor expression by including the effects of the paralleled damping resistor R_{damp}:

$$Q \approx \frac{R_{in} \,||\, R_{damp}}{\sqrt{\frac{L}{C}}} = \frac{R_{in} R_{damp}}{(R_{damp} + R)\sqrt{\frac{L}{C}}} \quad (9.158)$$

If we extract the damping resistor from this equation, we obtain

$$R_{damp} = \frac{Q R_{in}\sqrt{\frac{L}{C}}}{R_{in} - Q\sqrt{\frac{L}{C}}} \quad (9.159)$$

To avoid oscillations, we can select a quality factor of 1 leading to a damping resistor of

$$R_{damp} = \frac{-63 \times \sqrt{\frac{150u}{10u}}}{-63 - \sqrt{\frac{150u}{10u}}} = 3.65\ \Omega \quad (9.160)$$

The dc-block capacitor is usually chosen to be 10 times the filter capacitor value. In our case, it corresponds to a 100-μF type. To test the damping effectiveness, we can first run an ac plot of the filter loaded by R_{in}. The simulation circuit appears in Figure 9.67, where all the elements have been installed. The negative resistance is emulated by a constant power source; see page 71 of [12]. The power is adjusted to 150 W while the dc rail is set to 100 V for the analysis.

The transfer function in Figure 9.68 shows how the added resistor effectively damps the filter. The capacitor can be selected to values different than 10 times the filter capacitor, but the damping efficiency then starts to weaken.

The various publications targeting on input filter problems have shown that having the filter output impedance well below the lowest locus of the converter

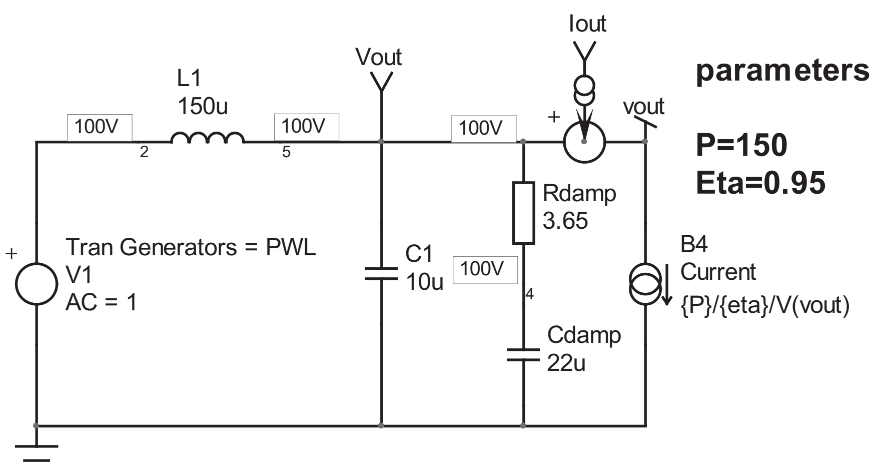

Figure 9.67 A constant-power source makes an ideal negative resistance load.

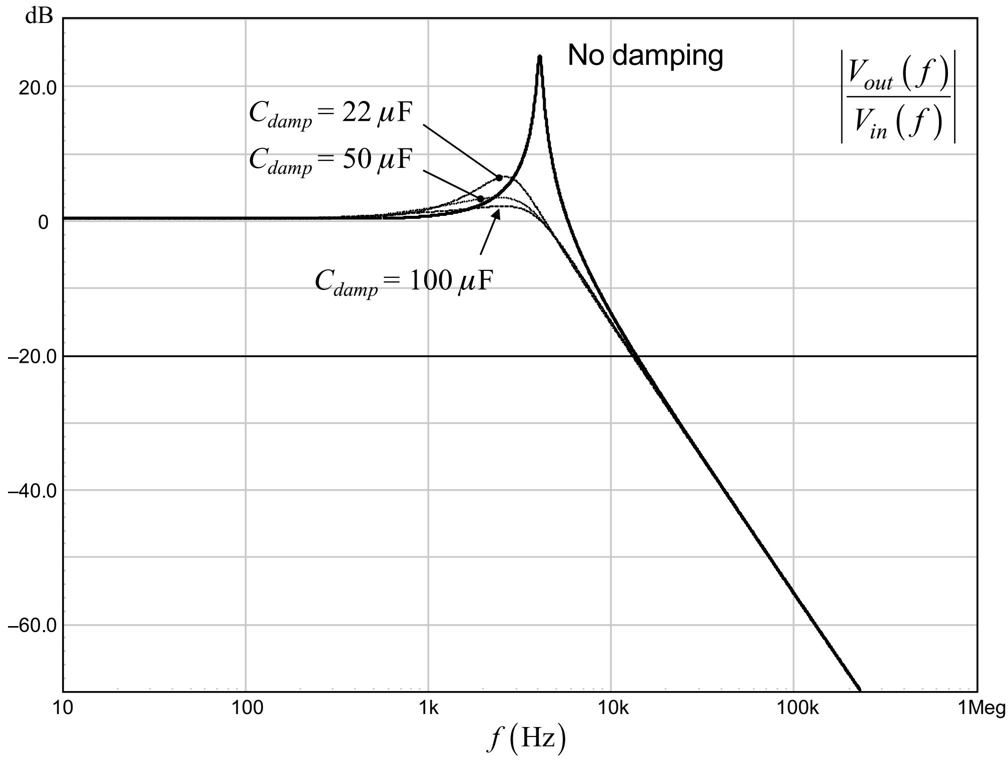

Figure 9.68 Without damping, you can clearly see the filter peaking. The resistor dissipates the ac energy and calms down any unwanted oscillations.

input impedance avoids potential instabilities. In other words, you have to damp your filter so that:

$$\left|Z_{out}(s)\right|_{\max} \ll \left|Z_{in}(s)\right|_{\min} \tag{9.161}$$

In this expression, the incremental resistance represents the dc value only. The complex input impedance depends on the closed-loop gain and must be carefully analyzed to check if (9.161) is satisfied along the frequency axis. Oscillations risks exist only up to the loop gain crossover frequency. Beyond that point, the negative incremental resistance reverts to a positive value as the converter is no longer able to correct the incoming high-frequency perturbations.

A simple check consists of plotting the damped filter output impedance and adding a line representing the negative incremental resistance defined by (9.157). This is the chart drawn in Figure 9.69. We have added the incremental resistance value, $20\log_{10}(63) = 36$ dBΩ as a horizontal reference. Without damping, the filter output impedance peaks so much that it exceeds the reference. Fortunately, once damping is added, a comfortable margin is created, making the design safe.

To further verify that the cure works as expected, we have stepped the input voltage in Figure 9.67 and checked the output voltage shape. This is what is shown in Figure 9.70 where the input voltage drops from 140 V to 100 V in 10 μs. When damping is adequate, the oscillations disappear immediately. If you reduce the

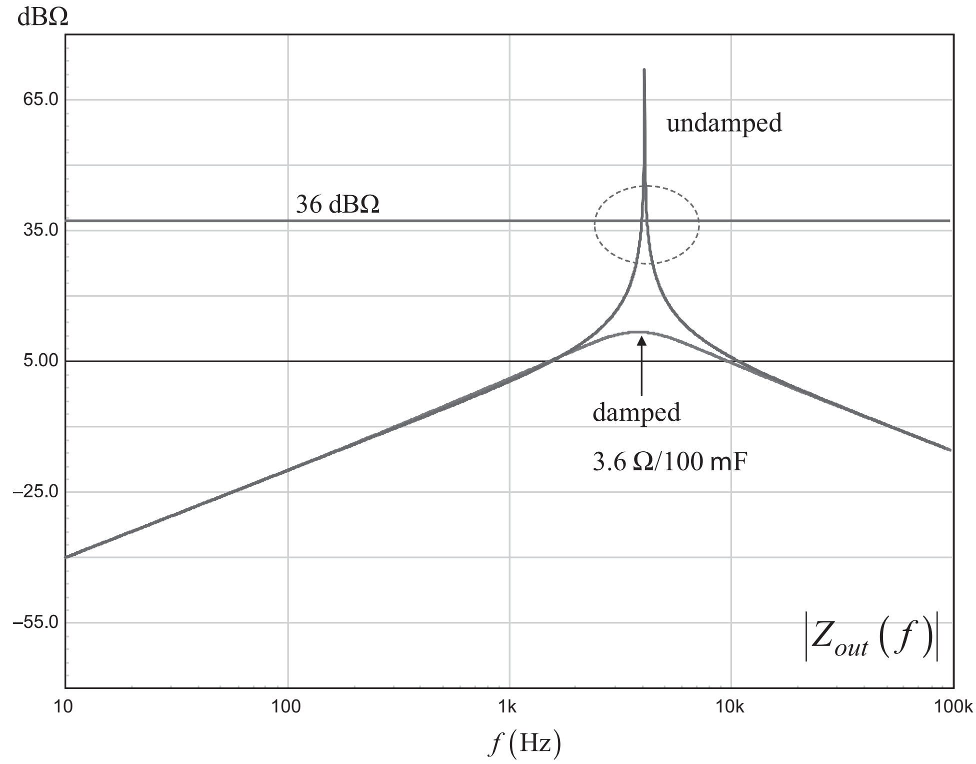

Figure 9.69 No overlap must exist between the incremental input resistance and filter output impedance.

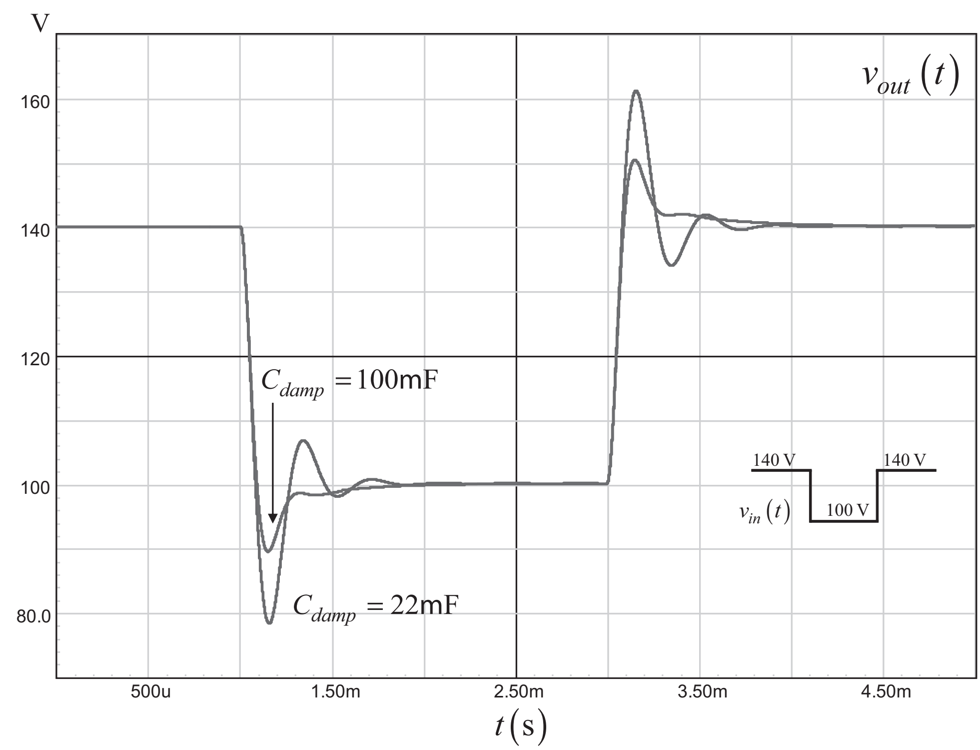

Figure 9.70 With a properly damped filter, oscillations are well damped and quickly die out.

100-μF capacitor to 22 μF, the damping becomes less efficient and a little ringing pops up.

The comprehensive analysis of such a converter will include the complete input impedance frequency sweep. More information on the subject will be found in [12, 16].

9.7 Conclusion

In this chapter, we learned how to measure a power converter loop gain. This stage is important, as it will let you know whether or not the assumptions you made during simulation or analytical analysis were confirmed by bench measurements. Some hidden parameters are difficult to predict, and these laboratories experiments are mandatory. At the beginning of the production, or during the pilot run, do not hesitate to sample some of the converters and recheck that stability is not jeopardized by components that were replaced at the last minute because of cheaper prices. You know that the ESR plays a role in the compensation scheme. Make sure high and low temperature transient responses are not too different; otherwise, this could indicate a parameter drift out of control. If you respect all these points and absolutely ban trial-and-error techniques, then you will be able to enjoy peaceful sleeps when mass production starts!

The design examples have shown that you can choose analytical analysis or simulation. Analytical analysis offers the advantage of unveiling the position of poles and zeros and how external parameters affect them. SPICE will give you the same response but without the insight brought by the analytical analysis. Combining both techniques, followed by real experiments, is the recipe for success.

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Conclusion

Writing a book on loop control is like a long, almost endless, journey: you know the destination but not all of the detours you are going to make before you reach it. The domain is so vast, and the exploration options so abundant, that to make one's way through all these meanders can quickly become an impossible exercise. By narrowing down the necessary knowledge to what is really important to our power conversion field, the work immediately becomes less complicated and no longer seemingly out of reach.

In this book, I have tried to give you the minimum set of necessary tools to either start designing a project or jump to a higher-level book. These tools, however, cannot be efficiently used if you do not understand the way they have been forged. Following the derivation lines not only helps you to progress in the field, but it also teaches the limits of the tools: in which cases you can use them and where it is necessary to select new ones. Therefore, as a recommendation, go through the lines I have derived and figure out how to reach the result yourself. It is time consuming, but this is the best way to learn.

Despite the length of the effort, I have really enjoyed writing this document. Not only because I, too, made progress in the area of control systems, but also because I hope I have shed a modestly different light on the field. If you, my reader friends, also feel this way after reading this book, then I will have accomplished my goal.

Appendix

Table of Variables and Acronyms Used in the Book

arg	The argument of a complex expression
BCM	Borderline conduction mode (same as CrM) or boundary conduction mode
BIBO	Bounded input bounded output
χ	The characteristic equation of a closed-loop transfer function, chi
CCM	Continuous conduction mode
CL	Closed loop, T_{CL} is the closed-loop gain for instance
CrM	Critical conduction mode
CTR	Current transfer ratio for an optocoupler
D	The converter averaged duty ratio
$d(t)$	The converter instantaneous duty ratio
$D(s)$	The Laplace expression of a transfer function denominator
δ	The logarithmic decrement, delta
ESR	Equivalent series resistance, also noted r_C or r_L respectively for a capacitor and an inductor
ESL	Equivalent series inductance
ε	The error voltage, epsilon
η	The converter efficiency, eta
f_c	The crossover frequency where $\lvert T(f_c)\rvert = 1$ or 0 dB
F_{sw}	The switching frequency
$G(s)$	The compensator Laplace expression
Gf_c	The gain deficit (or excess) at the selected crossover frequency
φ_m	The phase margin read at f_c
g_m	The transconductance of an operational transconductance amplifier (OTA)
GM	The gain margin in an open-loop Bode plot representation
$H(s)$	The plant Laplace expression
I_C	The capacitor dc current (0 at steady state)
$i_C(t)$	The instantaneous capacitor current
$\hat{i}_C$	The capacitor small-signal current
$i_d(t)$	The instantaneous diode current
I_d	The diode dc current
I_{in}	The input dc current of a given converter
I_L	The inductor dc current
$i_L(t)$	The instantaneous inductor current

$\hat{i}_L$	The inductor small-signal current
I_{out}	The dc output current
$i_{out}(t)$	The instantaneous output current
$\hat{i}_{out}$	The small-signal outputcurrent
k_d	The derivative term in a PID compensator
k_i	The integral term in a PID compensator
k_p	The proportional term in a PID compensator
L_p	The primary inductance of a transformer (usually in a flyback converter)
$\mathcal{L}\{f(t)\}$	The Laplace transform of a time-domain function f
LHP	Refers to the Left Half-Plane of the Argand chart
LHPP	A pole located in the LHP
LHPZ	A zero located in LHP
LTI	Linear time invariant
MIMO	Multi-input multi-output
$N(s)$	The Laplace expression of a transfer function numerator
OL	Open loop, T_{OL} is the open-loop gain for instance
ω	The angular frequency in radians per seconds, rad/s
ω_n or ω_0	The natural angular frequency in radians per seconds, rad/s
ω_d	The damped angular frequency in radians per seconds, rad/s
ω_r or ω_M	The resonant angular frequency in radians per seconds, rad/s
P_{in}	The converter input power
P_{out}	The converter output power
PID	Proportional integral derivative
PI	Proportional integral
Q	The quality factor of a filter
$r_n(t)$	The natural or free response (the excitation signal is set to zero)
$r_f(t)$	The forced response (the initial conditions are identically zero)
r_C	The series resistance of the capacitor
r_L	The series resistance of the inductor
RMS	Root mean square
R_{sense} or R_i	The sense resistor in a current-mode controlled converter. Sometimes called the burden resistor.
RHP	Refers to the right half-plane of the Argand chart
RHPP	A pole located in the RHP
RHPZ	A zero located in the RHP
s	$s = \sigma + j\omega$
SISO	Single output single input
SMPS	Switch mode power supply
SPICE	Simulation Program with Integrated Circuit Emphasis
τ	The time constant, tau, in second
$T(s)$	The loop gain Laplace expression
T_{sw}	The switching period
$v_C(t)$	The instantaneous voltage across a capacitor
V_C	The capacitor dc voltage
$v_c(t)$	The instantaneous control voltage
V_c	The dc control voltage

$v_L(t)$	The instantaneous voltage across an inductor
V_L	The inductor dc voltage (0 at steady state)
ζ	The damping ratio, zeta

Numbers and Prefixes in Operations

I have purposely used SPICE prefixes in all the operations to avoid scientific notations in equations. There is no space inserted between the number and the prefix. The prefixes are as follows:

$$1f = 10^{-15}$$

$$1p = 10^{-12}$$

$$1n = 10^{-9}$$

$$1u = 10^{-6}$$

$$1m = 10^{-3}$$

$$1k = 10^{3}$$

$$1Meg = 10^{6}$$

For example:

$$R_1 = 10\ \text{k}\Omega \quad C_1 = 3\ \mu\text{F}$$

$$\tau = R_1C_1 = 10k \times 3u = 30m = 0.03\ \text{s or } 30\ \text{ms}$$

About the Author

Christophe Basso is an application engineering director at ON Semiconductor in Toulouse, France. He has originated numerous integrated circuits, among which the NCP120X series has set new standards for low standby-power converters. SPICE simulation is one of his favorite subjects, and he authored two books on the subject. In his work, he promotes the combined usage of SPICE as a design companion, which, when associated with an equation-based approach, helps to understand how complex circuitries operate. This technique is appreciated and recognized by the numerous customers he visits worldwide. Developing new integrated circuits while helping and teaching design engineers is part of his professional activity in the field of ac-dc power conversion for the past 15 years.

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